Business Case: Delhivery - Feature Engineering

Colab link: https://colab.research.google.com/drive/1OnPKP2caX9C1UOP0G0AzD_9DbTQOvVJt?usp=sharing

About Delhivery

Delhivery is the largest and fastest-growing fully integrated player in India by revenue in Fiscal 2021. They aim to build the operating system for commerce, through a combination of world-class infrastructure, logistics operations of the highest quality, and cutting-edge engineering and technology capabilities.

What is expected?

The company wants to understand and process the data coming out of data engineering pipelines:

• Clean, sanitize and manipulate data to get useful features out of raw fields

• Make sense out of the raw data and help the data science team to build forecasting models on it

Importing Required Libraries

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib as mpl
import matplotlib.pyplot as plt
import scipy.stats as spy
```

Loading the Dataset and make copy

```
data = pd.read_csv('https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/001/551/original/delhivery_data.csv?1642751181')
# Copy of the above dataset
df = data.copy()
df.head()
```

	data	trip_creation_time	route_schedule_uuid	route_type	trip_uuid	source_center	source_name	destination_cente
0	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
1	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
2	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
3	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
4	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA

5 rows × 24 columns

```
# Shape of the dataset(Number of rows and columns)
df.shape
print('NUMBER OF ROWS IN DATASET:',df.shape[0])
print('NUMBER OF COLUMNS IN DATASET:',df.shape[1])
```

```
NUMBER OF ROWS IN DATASET: 144867
NUMBER OF COLUMNS IN DATASET: 24
```

```
# Total columns and rows in dataset:
df.size
```

```
3476808
```

```
# Columns Name:
df.columns
```

```
Index(['data', 'trip_creation_time', 'route_schedule_uuid', 'route_type',
       'trip_uuid', 'source_center', 'source_name', 'destination_center',
       'destination_name', 'od_start_time', 'od_end_time',
       'start_scan_to_end_scan', 'is_cutoff', 'cutoff_factor',
       'cutoff_timestamp', 'actual_distance_to_destination', 'actual_time',
       'osrm_time', 'osrm_distance', 'factor', 'segment_actual_time',
       'segment_osrm_time', 'segment_osrm_distance', 'segment_factor'],
      dtype='object')
```

```
# Datatyoes of the columns:
df.dtypes
```

```
data                              object
trip_creation_time                object
route_schedule_uuid               object
route_type                        object
trip_uuid                         object
source_center                     object
source_name                       object
destination_center                object
destination_name                  object
od_start_time                     object
od_end_time                       object
start_scan_to_end_scan           float64
is_cutoff                           bool
cutoff_factor                      int64
cutoff_timestamp                  object
actual_distance_to_destination   float64
actual_time                      float64
osrm_time                        float64
osrm_distance                    float64
factor                           float64
segment_actual_time              float64
segment_osrm_time                float64
segment_osrm_distance            float64
segment_factor                   float64
dtype: object
```

```
# Basic info abut the dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144867 entries, 0 to 144866
Data columns (total 24 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   data                            144867 non-null  object
 1   trip_creation_time              144867 non-null  object
 2   route_schedule_uuid             144867 non-null  object
 3   route_type                      144867 non-null  object
 4   trip_uuid                       144867 non-null  object
 5   source_center                   144867 non-null  object
 6   source_name                     144574 non-null  object
 7   destination_center              144867 non-null  object
 8   destination_name                144606 non-null  object
 9   od_start_time                   144867 non-null  object
 10  od_end_time                     144867 non-null  object
 11  start_scan_to_end_scan          144867 non-null  float64
 12  is_cutoff                       144867 non-null  bool
 13  cutoff_factor                   144867 non-null  int64
 14  cutoff_timestamp                144867 non-null  object
 15  actual_distance_to_destination  144867 non-null  float64
 16  actual_time                     144867 non-null  float64
 17  osrm_time                       144867 non-null  float64
 18  osrm_distance                   144867 non-null  float64
 19  factor                          144867 non-null  float64
 20  segment_actual_time             144867 non-null  float64
 21  segment_osrm_time               144867 non-null  float64
 22  segment_osrm_distance           144867 non-null  float64
```

```
 23  segment_factor                  144867 non-null  float64
dtypes: bool(1), float64(10), int64(1), object(12)
memory usage: 25.6+ MB
```

```
# Descriptive statistics of the data:
df.describe()
```

	start_scan_to_end_scan	cutoff_factor	actual_distance_to_destination	actual_time	osrm_time	osrm_distance	factor
count	144867.000000	144867.000000	144867.000000	144867.000000	144867.000000	144867.000000	144867.000000
mean	961.262986	232.926567	234.073372	416.927527	213.868272	284.771297	2.120107
std	1037.012769	344.755577	344.990009	598.103621	308.011085	421.119294	1.715421
min	20.000000	9.000000	9.000045	9.000000	6.000000	9.008200	0.144000
25%	161.000000	22.000000	23.355874	51.000000	27.000000	29.914700	1.604264
50%	449.000000	66.000000	66.126571	132.000000	64.000000	78.525800	1.857143
75%	1634.000000	286.000000	286.708875	513.000000	257.000000	343.193250	2.213483
max	7898.000000	1927.000000	1927.447705	4532.000000	1686.000000	2326.199100	77.387097

```
# Top 5 rows of the data:
df.head()
```

	data	trip_creation_time	route_schedule_uuid	route_type	trip_uuid	source_center	source_name	destination_cente
0	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
1	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
2	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
3	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
4	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA

5 rows × 24 columns

```
# Laset 5 rows of the data:
df.tail()
```

	data	trip_creation_time	route_schedule_uuid	route_type	trip_uuid	source_center	source_name	destination_cen
144862	training	2018-09-20 16:24:28.436231	thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5...	Carting	trip-153746066843555182	IND131028AAB	Sonipat_Kundli_H (Haryana)	IND000000/
144863	training	2018-09-20 16:24:28.436231	thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5...	Carting	trip-153746066843555182	IND131028AAB	Sonipat_Kundli_H (Haryana)	IND000000/
144864	training	2018-09-20 16:24:28.436231	thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5...	Carting	trip-153746066843555182	IND131028AAB	Sonipat_Kundli_H (Haryana)	IND000000/
144865	training	2018-09-20 16:24:28.436231	thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5...	Carting	trip-153746066843555182	IND131028AAB	Sonipat_Kundli_H (Haryana)	IND000000/
144866	training	2018-09-20 16:24:28.436231	thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5...	Carting	trip-153746066843555182	IND131028AAB	Sonipat_Kundli_H (Haryana)	IND000000/

5 rows × 24 columns

Basic Analysis:

```
# Drop the not necessary columns(or unknown fields)
unknown_fields = ['is_cutoff', 'cutoff_factor', 'cutoff_timestamp', 'factor', 'segment_factor']
df = df.drop(columns = unknown_fields)
```

```
# Number of uniques entities in each column
for i in df.columns:
  print(f'Uniques entity for column "{i}" = {df[i].nunique()}')
```

```
Uniques entity for column "data" = 2
Uniques entity for column "trip_creation_time" = 14817
Uniques entity for column "route_schedule_uuid" = 1504
Uniques entity for column "route_type" = 2
Uniques entity for column "trip_uuid" = 14817
Uniques entity for column "source_center" = 1508
Uniques entity for column "source_name" = 1498
Uniques entity for column "destination_center" = 1481
Uniques entity for column "destination_name" = 1468
Uniques entity for column "od_start_time" = 26369
Uniques entity for column "od_end_time" = 26369
Uniques entity for column "start_scan_to_end_scan" = 1915
Uniques entity for column "actual_distance_to_destination" = 144515
Uniques entity for column "actual_time" = 3182
Uniques entity for column "osrm_time" = 1531
Uniques entity for column "osrm_distance" = 138046
Uniques entity for column "segment_actual_time" = 747
Uniques entity for column "segment_osrm_time" = 214
Uniques entity for column "segment_osrm_distance" = 113799
```

```
# Data type conversion:(object to category)
df['data'] = df['data'].astype('category')
df['route_type'] = df['route_type'].astype('category')
```

Converting time columns into pandas datetime.

```
# Updataing the datatype of the datetime columns from object
datetime_col= ['trip_creation_time', 'od_start_time', 'od_end_time']
for i in datetime_col:
  df[i] = pd.to_datetime(df[i])
```

```
# The memory usage of each column in bytes
df.memory_usage()
```

```
Index                                 128
data                               144991
trip_creation_time                1158936
route_schedule_uuid               1158936
route_type                         144991
trip_uuid                         1158936
source_center                     1158936
source_name                       1158936
destination_center                1158936
destination_name                  1158936
od_start_time                     1158936
od_end_time                       1158936
start_scan_to_end_scan            1158936
actual_distance_to_destination    1158936
actual_time                       1158936
osrm_time                         1158936
osrm_distance                     1158936
segment_actual_time               1158936
segment_osrm_time                 1158936
segment_osrm_distance             1158936
dtype: int64
```

```
float_col = ['actual_distance_to_destination', 'actual_time', 'osrm_time', 'osrm_distance',
                     'segment_actual_time', 'segment_osrm_time', 'segment_osrm_distance']
for i in float_col:
    print(df[i].max())
```

```
1927.4477046975032
4532.0
```

```
1686.0
2326.1991000000003
3051.0
1611.0
2191.4037000000003
```

Therefore we can observe that max value in float column id less so we can convert it into float32.

```
for i in float_col:
  df[i] = df[i].astype('float32')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144867 entries, 0 to 144866
Data columns (total 19 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   data                            144867 non-null  category
 1   trip_creation_time              144867 non-null  datetime64[ns]
 2   route_schedule_uuid             144867 non-null  object
 3   route_type                      144867 non-null  category
 4   trip_uuid                       144867 non-null  object
 5   source_center                   144867 non-null  object
 6   source_name                     144574 non-null  object
 7   destination_center              144867 non-null  object
 8   destination_name                144606 non-null  object
 9   od_start_time                   144867 non-null  datetime64[ns]
 10  od_end_time                     144867 non-null  datetime64[ns]
 11  start_scan_to_end_scan          144867 non-null  float64
 12  actual_distance_to_destination  144867 non-null  float32
 13  actual_time                     144867 non-null  float32
 14  osrm_time                       144867 non-null  float32
 15  osrm_distance                   144867 non-null  float32
 16  segment_actual_time             144867 non-null  float32
 17  segment_osrm_time               144867 non-null  float32
 18  segment_osrm_distance           144867 non-null  float32
dtypes: category(2), datetime64[ns](3), float32(7), float64(1), object(6)
memory usage: 15.2+ MB
```

So memory reduction is now 15.2+ MB which was earlier 25.6+MB, making it comparetively slower.

```
# Time Period for which the data is given:
print(f"start time : { df['od_start_time'].min()}")
print(f"End time : { df['od_end_time'].max()}")
```

```
start time : 2018-09-12 00:00:16.535741
End time : 2018-10-08 03:00:24.353479
```

Handle missing values in the data.

```
# Check for missing data in each columns:
print(f'If any null values present in the dataset : {np.any(df.isna())}')
print(f'Total Number of missing data in the dataset is : {df.isna().sum().sum()}')
df.isna().sum()
```

```
If any null values present in the dataset : True
Total Number of missing data in the dataset is : 554
data                              0
trip_creation_time                0
route_schedule_uuid               0
route_type                        0
trip_uuid                         0
source_center                     0
source_name                     293
destination_center                0
destination_name                261
od_start_time                     0
od_end_time                       0
start_scan_to_end_scan            0
actual_distance_to_destination    0
actual_time                       0
osrm_time                         0
osrm_distance                     0
segment_actual_time               0
```

```
segment_osrm_time                  0
segment_osrm_distance              0
dtype: int64
```

```
# lets find unique values in the 'source_center' column of the DataFrame 'df' for rows where the 'source_name' value is null.
```

```
# We have only two columns with null values: source_name and destination_name
missing_source_name = df.loc[df['source_name'].isnull(), 'source_center'].unique()
missing_source_name
```

```
array(['IND342902A1B', 'IND577116AAA', 'IND282002AAD', 'IND465333A1B',
       'IND841301AAC', 'IND509103AAC', 'IND126116AAA', 'IND331022A1B',
       'IND505326AAB', 'IND852118A1B'], dtype=object)
```

```
for i in missing_source_name:
    unique_source_name = df.loc[df['source_center'] == i, 'source_name'].unique()
    if pd.isna(unique_source_name):
        print("Source Center :", i, "-" * 10, "Source Name :", 'Not Found')
    else :
        print("Source Center :", i, "-" * 10, "Source Name :", unique_source_name)
```

```
Source Center : IND342902A1B ---------- Source Name : Not Found
Source Center : IND577116AAA ---------- Source Name : Not Found
Source Center : IND282002AAD ---------- Source Name : Not Found
Source Center : IND465333A1B ---------- Source Name : Not Found
Source Center : IND841301AAC ---------- Source Name : Not Found
Source Center : IND509103AAC ---------- Source Name : Not Found
Source Center : IND126116AAA ---------- Source Name : Not Found
Source Center : IND331022A1B ---------- Source Name : Not Found
Source Center : IND505326AAB ---------- Source Name : Not Found
Source Center : IND852118A1B ---------- Source Name : Not Found
```

```
missing_destination_name = df.loc[df['destination_name'].isnull(), 'destination_center'].unique()
missing_destination_name
```

```
array(['IND342902A1B', 'IND577116AAA', 'IND282002AAD', 'IND465333A1B',
       'IND841301AAC', 'IND505326AAB', 'IND852118A1B', 'IND126116AAA',
       'IND509103AAC', 'IND221005A1A', 'IND250002AAC', 'IND331001A1C',
       'IND122015AAC'], dtype=object)
```

```
for i in missing_destination_name:
    unique_destination_name = df.loc[df['destination_center'] == i, 'destination_name'].unique()
    if (pd.isna(unique_destination_name)) or (unique_destination_name.size == 0):
        print("Destination Center :", i, "-" * 10, "Destination Name :", 'Not Found')
    else :
        print("Destination Center :", i, "-" * 10, "Destination Name :", unique_destination_name)
```

```
Destination Center : IND342902A1B ---------- Destination Name : Not Found
Destination Center : IND577116AAA ---------- Destination Name : Not Found
Destination Center : IND282002AAD ---------- Destination Name : Not Found
Destination Center : IND465333A1B ---------- Destination Name : Not Found
Destination Center : IND841301AAC ---------- Destination Name : Not Found
Destination Center : IND505326AAB ---------- Destination Name : Not Found
Destination Center : IND852118A1B ---------- Destination Name : Not Found
Destination Center : IND126116AAA ---------- Destination Name : Not Found
Destination Center : IND509103AAC ---------- Destination Name : Not Found
Destination Center : IND221005A1A ---------- Destination Name : Not Found
Destination Center : IND250002AAC ---------- Destination Name : Not Found
Destination Center : IND331001A1C ---------- Destination Name : Not Found
Destination Center : IND122015AAC ---------- Destination Name : Not Found
```

```
# Rows where 'source_name' is null
rows_with_null_source_name = df['source_name'].isnull()

# Get 'source_center' values from these rows
source_centers_with_null_source_name = df.loc[rows_with_null_source_name, 'source_center']

# Check if each of these 'source_center' values is in 'missing_destination_name'
all_values_in_missing_destination_name = np.all(source_centers_with_null_source_name.isin(missing_destination_name))

all_values_in_missing_destination_name
```

```
False
```

Treating missing 'destination names' and 'source names'

```
# Initialize a counter for unique placeholder names
count = 1

# Impute missing 'destination_name' values
for i in missing_destination_name:
    placeholder_name = f'location_{count}'
    df.loc[df['destination_center'] == i, 'destination_name'] = df.loc[df['destination_center'] == i, 'destination_name'].replace(np.nan, pl
    count += 1

# Create and apply a mapping for 'source_center' to 'destination_name'
d = {}
for i in missing_source_name:
    unique_names = df.loc[df['destination_center'] == i, 'destination_name'].unique()
    d[i] = unique_names if len(unique_names) > 0 else [f'location_{count}']
    count += (len(unique_names) == 0)

# Simplify the mapping to use only the first 'destination_name' or the assigned placeholder
d2 = {idx: val[0] for idx, val in d.items()}

# Apply the mapping to fill missing 'source_name' values
for i, v in d2.items():
    df.loc[df['source_center'] == i, 'source_name'] = df.loc[df['source_center'] == i, 'source_name'].replace(np.nan, v)
```

```
df.isna().sum()
```

```
data                             0
trip_creation_time               0
route_schedule_uuid              0
route_type                       0
trip_uuid                        0
source_center                    0
source_name                      0
destination_center               0
destination_name                 0
od_start_time                    0
od_end_time                      0
start_scan_to_end_scan           0
actual_distance_to_destination   0
actual_time                      0
osrm_time                        0
osrm_distance                    0
segment_actual_time              0
segment_osrm_time                0
segment_osrm_distance            0
dtype: int64
```

```
df.describe(include = 'object')
```

	route_schedule_uuid	trip_uuid	source_center	source_name	destination_center	destination_name
count	144867	144867	144867	144867	144867	144867
unique	1504	14817	1508	1508	1481	1481
top	thanos::sroute:4029a8a2-6c74-4b7e-a6d8-f9e069f...	trip-153811219535896559	IND000000ACB	Gurgaon_Bilaspur_HB (Haryana)	IND000000ACB	Gurgaon_Bilaspur_HB (Haryana)
freq	1812	101	23347	23347	15192	15192

Merging the rows

1. Grouping by segment

a. Create a unique identifier for different segments of a trip based on the combination of the trip_uuid, source_center, and destination_center and name it as segment_key.

```
df.columns
```

```
Index(['data', 'trip_creation_time', 'route_schedule_uuid', 'route_type',
       'trip_uuid', 'source_center', 'source_name', 'destination_center',
```

```
       'destination_name', 'od_start_time', 'od_end_time',
       'start_scan_to_end_scan', 'actual_distance_to_destination',
       'actual_time', 'osrm_time', 'osrm_distance', 'segment_actual_time',
       'segment_osrm_time', 'segment_osrm_distance'],
      dtype='object')
```

```
# Creating a unique identifier for each segment of a trip
df['segment_key'] = df['trip_uuid'] + '_' + df['source_center'] + '_' + df['destination_center']

# Using cumsum() to merge the rows for specified columns based on segment_key
cumulative_columns = {
    'segment_actual_time': 'segment_actual_time_sum',
    'segment_osrm_distance': 'segment_osrm_distance_sum',
    'segment_osrm_time': 'segment_osrm_time_sum'
}

for original_col, new_col in cumulative_columns.items():
    df[new_col] = df.groupby('segment_key')[original_col].cumsum()

df[['segment_key', 'segment_actual_time', 'segment_actual_time_sum',
      'segment_osrm_distance', 'segment_osrm_distance_sum',
      'segment_osrm_time', 'segment_osrm_time_sum']]
```

	segment_key	segment_actual_time	segment_actual_time_sum	segment_osrm_distance	segment_osr
0	trip-153741093647649320_IND388121AAA_IND388620AAB	14.0	14.0	11.965300	
1	trip-153741093647649320_IND388121AAA_IND388620AAB	10.0	24.0	9.759000	
2	trip-153741093647649320_IND388121AAA_IND388620AAB	16.0	40.0	10.815200	
3	trip-153741093647649320_IND388121AAA_IND388620AAB	21.0	61.0	13.022400	
4	trip-153741093647649320_IND388121AAA_IND388620AAB	6.0	67.0	3.915300	
...	...	...	...	...	
144862	trip-153746066843555182_IND131028AAB_IND000000ACB	12.0	92.0	8.185800	
144863	trip-153746066843555182_IND131028AAB_IND000000ACB	26.0	118.0	17.372499	
144864	trip-153746066843555182_IND131028AAB_IND000000ACB	20.0	138.0	20.705299	
144865	trip-153746066843555182_IND131028AAB_IND000000ACB	17.0	155.0	18.888500	
144866	trip-153746066843555182_IND131028AAB_IND000000ACB	268.0	423.0	8.808800	

144867 rows × 7 columns

```
df.head()
```

	data	trip_creation_time	route_schedule_uuid	route_type	trip_uuid	source_center	source_name	destination_cente
0	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
1	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
2	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
3	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA
4	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AA

5 rows × 23 columns

```
# Aggregating at segment level

# Creating a dictionary for aggregation at segment level
create_segment_dict = {
  'trip_uuid' : 'first',
  'data': 'first',
  'route_type': 'first',
  'trip_creation_time': 'first',
  'source_name': 'first',
  'destination_name': 'last',
  'od_start_time': 'first',
  'od_end_time': 'last',
  'start_scan_to_end_scan': 'first',
  'actual_distance_to_destination': 'last',
  'actual_time': 'last',
  'osrm_time': 'last',
  'osrm_distance': 'last',
  'segment_actual_time' : 'sum',
  'segment_osrm_time' : 'sum',
  'segment_osrm_distance' : 'sum',
  'segment_actual_time_sum': 'last',
  'segment_osrm_time_sum': 'last',
  'segment_osrm_distance_sum': 'last',
}

# Grouping by segment_key and applying the aggregation operations
seg_agg_data = df.groupby('segment_key').agg(create_segment_dict).reset_index()

# Sorting in ascending order
seg_agg_data = seg_agg_data.sort_values(by=['segment_key','od_end_time'])

seg_agg_data
```

	segment_key	trip_uuid	data	route_type	trip_creation_time	source_nam
0	trip-153671041653548748_IND209304AAA_IND000000ACB	trip-153671041653548748	training	FTL	2018-09-12 00:00:16.535741	Kanpur_Central_H_((Uttar Pradesh
1	trip-153671041653548748_IND462022AAA_IND209304AAA	trip-153671041653548748	training	FTL	2018-09-12 00:00:16.535741	Bhopal_Trnsport_H (Madhya Pradesh
2	trip-153671042288605164_IND561203AAB_IND562101AAA	trip-153671042288605164	training	Carting	2018-09-12 00:00:22.886430	Doddablpur_ChikaDPP_D (Karnataka
3	trip-153671042288605164_IND572101AAA_IND561203AAB	trip-153671042288605164	training	Carting	2018-09-12 00:00:22.886430	Tumkur_Veersagr_ (Karnataka
4	trip-153671043369099517_IND000000ACB_IND160002AAC	trip-153671043369099517	training	FTL	2018-09-12 00:00:33.691250	Gurgaon_Bilaspur_HB (Haryana
...	...	...	...	...	...	..
26363	trip-153861115439069069_IND628204AAA_IND627657AAA	trip-153861115439069069	test	Carting	2018-10-03 23:59:14.390954	Tirchchndr_Shnmgprm_D (Tamil Nadu
26364	trip-153861115439069069_IND628613AAA_IND627005AAA	trip-153861115439069069	test	Carting	2018-10-03 23:59:14.390954	Peikulam_SriVnktpm_D (Tamil Nadu
26365	trip-153861115439069069_IND628801AAA_IND628204AAA	trip-153861115439069069	test	Carting	2018-10-03 23:59:14.390954	Eral_Busstand_D (Tam Nadu
26366	trip-153861118270144424_IND583119AAA_IND583101AAA	trip-153861118270144424	test	FTL	2018-10-03 23:59:42.701692	Sandur_WrdN1DPP_D (Karnataka
26367	trip-153861118270144424_IND583201AAA_IND583119AAA	trip-153861118270144424	test	FTL	2018-10-03 23:59:42.701692	Hospet (Karnataka

26368 rows × 20 columns

The rows have been merged based on the unique segment_key, which is a combination of trip_uuid, source_center, and destination_center. The aggregated dataset reflects the total values for each segment of the trip. How the aggregation was performed:

Numerical Fields:

- The Feilds segment_actual_time, segment_osrm_time, segment_osrm_distance were summed up.
- This gives a total measure of time and distance for each unique segment.

Categorical/Boolean Fields:

- For fields like route_type, the first value in each group was kept.
- The data field, which distinguishes between training and testing data, was also preserved with the first value.

Source and Destination Names:

- The source_name field retains the first source name for each segment.
- The destination_name field holds the last destination name for each segment.

Feature Engineering:

1. Calculate time taken between od_start_time and od_end_time and keep it as a feature named od_time_diff_hour. Drop the original columns, if required.

```
# Time difference between od_start_time and od_end_time
seg_agg_data['od_total_time'] = (
    seg_agg_data['od_end_time'] - seg_agg_data['od_start_time']
).dt.total_seconds() / 60
```

Creating new feature

```
# Performing aggregation on the groupby trip uuid and name it as fe(feature_engineering)

# All the aggregate that we will perform on the different columns
create_trip_dict={
  'data' : 'first',
  'route_type' : 'first',
  'od_total_time' : 'sum',
  'trip_creation_time' : 'first'
```

```
  'trip_creation_time' : 'first',
  'start_scan_to_end_scan' : 'sum',
  'source_name': 'first',
  'destination_name': 'last',
  'actual_distance_to_destination' : 'sum',
  'actual_time' : 'sum',
  'osrm_time' : 'sum',
  'osrm_distance' : 'sum',
  'segment_actual_time': 'sum',
  'segment_osrm_time': 'sum',
  'segment_osrm_distance': 'sum',
  'segment_actual_time_sum': 'sum',
  'segment_osrm_time_sum': 'sum',
  'segment_osrm_distance_sum': 'sum',
  }

# Aggregation on diff col
df1 = seg_agg_data.groupby('trip_uuid').agg(create_trip_dict).reset_index()
df1
```

	trip_uuid	data	route_type	od_total_time	trip_creation_time	start_scan_to_end_scan	source_name	d
0	trip-153671041653548748	training	FTL	2260.109800	2018-09-12 00:00:16.535741	2259.0	Kanpur_Central_H_6 (Uttar Pradesh)	Ka
1	trip-153671042288605164	training	Carting	181.611874	2018-09-12 00:00:22.886430	180.0	Doddablpur_ChikaDPP_D (Karnataka)	Doddab
2	trip-153671043369099517	training	FTL	3934.362520	2018-09-12 00:00:33.691250	3933.0	Gurgaon_Bilaspur_HB (Haryana)	Gur
3	trip-153671046011330457	training	Carting	100.494935	2018-09-12 00:01:00.113710	100.0	Mumbai Hub (Maharashtra)	M
4	trip-153671052974046625	training	FTL	718.349042	2018-09-12 00:02:09.740725	717.0	Bellary_Dc (Karnataka)	San
...	...	...	...	...	...	...	...	
14812	trip-153861095625827784	test	Carting	258.028928	2018-10-03 23:55:56.258533	257.0	Chandigarh_Mehmdpur_H (Punjab)	Chandig
14813	trip-153861104386292051	test	Carting	60.590521	2018-10-03 23:57:23.863155	60.0	FBD_Balabhgarh_DPC (Haryana)	Farid
14814	trip-153861106442901555	test	Carting	422.119867	2018-10-03 23:57:44.429324	421.0	Kanpur_GovndNgr_DC (Uttar Pradesh)	Kanp
14815	trip-153861115439069069	test	Carting	348.512862	2018-10-03 23:59:14.390954	347.0	Tirunelveli_VdkkuSrt_I (Tamil Nadu)	Tirchch
14816	trip-153861118270144424	test	FTL	354.407571	2018-10-03 23:59:42.701692	353.0	Sandur_WrdN1DPP_D (Karnataka)	San

14817 rows × 18 columns

Build some features to prepare the data for actual analysis. Extract features from the below fields:

1. Destination Name: Split and extract features out of destination. City-place-code (State)
2. Source Name: Split and extract features out of destination. City-place-code (State)
3. Trip_creation_time: Extract features like month, year and day etc

```
def location_name_to_state(x):
    l = x.split('(')
    if len(l) == 1:
        return l[0]
    else:
        return l[1].replace(')', "")

def location_name_to_city(x):
    if 'location' in x:
        return 'unknown_city'
    else:
        l = x.split()[0].split('_')
        if 'CCU' in x:
            return 'Kolkata'
        elif 'MAA' in x.upper():
            return 'Chennai'
        elif ('HBR' in x.upper()) or ('BLR' in x.upper()):
            return 'Bengaluru'
        elif 'FBD' in x.upper():
            return 'Faridabad'
        elif 'BOM' in x.upper():
            return 'Mumbai'
        elif 'DEL' in x.upper():
            return 'Delhi'
        elif 'OK' in x.upper():
            return 'Delhi'
        elif 'GZB' in x.upper():
            return 'Ghaziabad'
        elif 'GGN' in x.upper():
            return 'Gurgaon'
        elif 'AMD' in x.upper():
            return 'Ahmedabad'
        elif 'CJB' in x.upper():
            return 'Coimbatore'
        elif 'HYD' in x.upper():
            return 'Hyderabad'
        return l[0]

def location_name_to_place(x):
    if 'location' in x:
        return x
    elif 'HBR' in x:
        return 'HBR Layout PC'
    else:
        l = x.split()[0].split('_', 1)
        if len(l) == 1:
            return 'unknown_place'
        else:
            return l[1]
```

```
# Extract state city place code from source name:

df1['source_state'] = df1['source_name'].apply(location_name_to_state)
df1['source_city'] = df1['source_name'].apply(location_name_to_city)
df1['source_place'] = df1['source_name'].apply(location_name_to_place)
```

```
# Extract state city place code from destination name:

df1['destination_state'] = df1['destination_name'].apply(location_name_to_state)
df1['destination_city'] = df1['destination_name'].apply(location_name_to_city)
df1['destination_place'] = df1['destination_name'].apply(location_name_to_place)
```

```
df1.head(2)
print(df.shape)
```

```
(144867, 23)
```

```
# Droping the source name and destination name columns from df1 (dataframe):
df1.drop(['source_name','destination_name'],axis = 1,inplace = True)
```

```
df1.head(2)
```

	trip_uuid	data	route_type	od_total_time	trip_creation_time	start_scan_to_end_scan	actual_distance_to_destination	ac
0	trip-153671041653548748	training	FTL	2260.109800	2018-09-12 00:00:16.535741	2259.0	824.732849	
1	trip-153671042288605164	training	Carting	181.611874	2018-09-12 00:00:22.886430	180.0	73.186905	

2 rows × 22 columns

Trip_creation_time: Extract features like month, year and day etc:

As out column trip creation time is already in datatime datatype then we can easily extration out the month year day..etc fields from it.

```
df1['trip_creation_month'] = df1['trip_creation_time'].dt.month
df1['trip_creation_month'] = df1['trip_creation_month'].astype('int8')
df1['trip_creation_year'] = df1['trip_creation_time'].dt.year
df1['trip_creation_year'] = df1['trip_creation_year'].astype('int16')
df1['trip_creation_day'] = df1['trip_creation_time'].dt.day
df1['trip_creation_day'] = df1['trip_creation_day'].astype('int8')
df1['trip_creation_hour'] = df1['trip_creation_time'].dt.hour
df1['trip_creation_hour'] = df1['trip_creation_hour'].astype('int8')
df1['trip_creation_week'] = df1['trip_creation_time'].dt.isocalendar().week
df1['trip_creation_week'] = df1['trip_creation_week'].astype('int8')
df1['trip_creation_weekday'] = df1['trip_creation_time'].dt.day_of_week
df1['trip_creation_weekday'] = df1['trip_creation_weekday'].astype('int8')
```

Now lets see the dataset basic info:

```
df1.head()
```

	trip_uuid	data	route_type	od_total_time	trip_creation_time	start_scan_to_end_scan	actual_distance_to_destination	ac
0	trip-153671041653548748	training	FTL	2260.109800	2018-09-12 00:00:16.535741	2259.0	824.732849	
1	trip-153671042288605164	training	Carting	181.611874	2018-09-12 00:00:22.886430	180.0	73.186905	
2	trip-153671043369099517	training	FTL	3934.362520	2018-09-12 00:00:33.691250	3933.0	1927.404297	
3	trip-153671046011330457	training	Carting	100.494935	2018-09-12 00:01:00.113710	100.0	17.175274	
4	trip-153671052974046625	training	FTL	718.349042	2018-09-12 00:02:09.740725	717.0	127.448502	

5 rows × 28 columns

```
df1.shape
```

```
(14817, 28)
```

```
df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14817 entries, 0 to 14816
Data columns (total 28 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   trip_uuid                       14817 non-null  object
 1   data                            14817 non-null  category
 2   route_type                      14817 non-null  category
 3   od_total_time                   14817 non-null  float64
 4   trip_creation_time              14817 non-null  datetime64[ns]
 5   start_scan_to_end_scan          14817 non-null  float64
 6   actual_distance_to_destination  14817 non-null  float32
 7   actual_time                     14817 non-null  float32
 8   osrm_time                       14817 non-null  float32
 9   osrm_distance                   14817 non-null  float32
 10  segment_actual_time             14817 non-null  float32
 11  segment_osrm_time               14817 non-null  float32
```

```
 12  segment_osrm_distance           14817 non-null  float32
 13  segment_actual_time_sum         14817 non-null  float32
 14  segment_osrm_time_sum           14817 non-null  float32
 15  segment_osrm_distance_sum       14817 non-null  float32
 16  source_state                    14817 non-null  object
 17  source_city                     14817 non-null  object
 18  source_place                    14817 non-null  object
 19  destination_state               14817 non-null  object
 20  destination_city                14817 non-null  object
 21  destination_place               14817 non-null  object
 22  trip_creation_month             14817 non-null  int8
 23  trip_creation_year              14817 non-null  int16
 24  trip_creation_day               14817 non-null  int8
 25  trip_creation_hour              14817 non-null  int8
 26  trip_creation_week              14817 non-null  int8
 27  trip_creation_weekday           14817 non-null  int8
dtypes: category(2), datetime64[ns](1), float32(10), float64(2), int16(1), int8(5), object(7)
memory usage: 1.8+ MB
```

```
df1.describe().T
```

	count	mean	std	min	25%	50%	75%	max
od_total_time	14817.0	531.795209	659.184525	23.461468	149.930591	280.765626	638.199071	7898.551955
start_scan_to_end_scan	14817.0	530.810016	658.705957	23.000000	149.000000	280.000000	637.000000	7898.000000
actual_distance_to_destination	14817.0	164.477829	305.388153	9.002461	22.837238	48.474072	164.583206	2186.531738
actual_time	14817.0	357.143768	561.396118	9.000000	67.000000	149.000000	370.000000	6265.000000
osrm_time	14817.0	161.384018	271.360992	6.000000	29.000000	60.000000	168.000000	2032.000000
osrm_distance	14817.0	204.344711	370.395569	9.072900	30.819201	65.618805	208.475006	2840.081055
segment_actual_time	14817.0	353.892273	556.247925	9.000000	66.000000	147.000000	367.000000	6230.000000
segment_osrm_time	14817.0	180.949783	314.542053	6.000000	31.000000	65.000000	185.000000	2564.000000
segment_osrm_distance	14817.0	223.201157	416.628387	9.072900	32.654499	70.154404	218.802399	3523.632324
segment_actual_time_sum	14817.0	353.892273	556.247925	9.000000	66.000000	147.000000	367.000000	6230.000000
segment_osrm_time_sum	14817.0	180.949783	314.542053	6.000000	31.000000	65.000000	185.000000	2564.000000
segment_osrm_distance_sum	14817.0	223.201157	416.628387	9.072900	32.654499	70.154404	218.802399	3523.632324
trip_creation_month	14817.0	9.120672	0.325757	9.000000	9.000000	9.000000	9.000000	10.000000
trip_creation_year	14817.0	2018.000000	0.000000	2018.000000	2018.000000	2018.000000	2018.000000	2018.000000
trip_creation_day	14817.0	18.370790	7.893275	1.000000	14.000000	19.000000	25.000000	30.000000
trip_creation_hour	14817.0	12.449821	7.986553	0.000000	4.000000	14.000000	20.000000	23.000000
trip_creation_week	14817.0	38.295944	0.967872	37.000000	38.000000	38.000000	39.000000	40.000000
trip_creation_weekday	14817.0	2.919349	1.927769	0.000000	1.000000	3.000000	5.000000	6.000000

```
df1.describe(include = 'object').T
```

	count	unique	top	freq
trip_uuid	14817	14817	trip-153671041653548748	1
source_state	14817	34	Maharashtra	2714
source_city	14817	690	Mumbai	1442
source_place	14817	761	Bilaspur_HB	1063
destination_state	14817	39	Maharashtra	2561
destination_city	14817	806	Mumbai	1548
destination_place	14817	850	Bilaspur_HB	821

In-depth analysis:

```
df1.columns
```

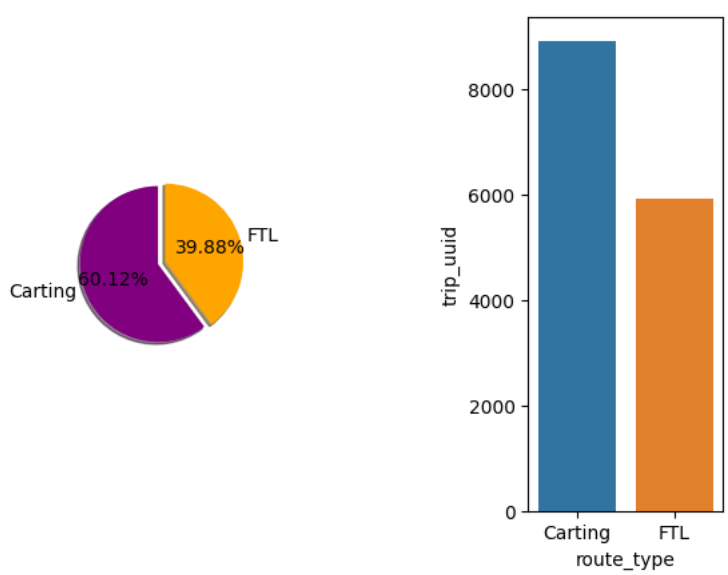
```
Index(['trip_uuid', 'data', 'route_type', 'od_total_time',
       'trip_creation_time', 'start_scan_to_end_scan',
       'actual_distance_to_destination', 'actual_time', 'osrm_time',
       'osrm_distance', 'segment_actual_time', 'segment_osrm_time',
       'segment_osrm_distance', 'segment_actual_time_sum',
       'segment_osrm_time_sum', 'segment_osrm_distance_sum', 'source_state',
       'source_city', 'source_place', 'destination_state', 'destination_city',
       'destination_place', 'trip_creation_month', 'trip_creation_year',
       'trip_creation_day', 'trip_creation_hour', 'trip_creation_week',
       'trip_creation_weekday'],
      dtype='object')
```

```
def addlabels(x,y):
    for i in range(len(x)):
        plt.text(i, y[i], y[i], ha = 'center')
```

```
# Distribution of the orders by route type
df_route = df1.groupby('route_type')['trip_uuid'].agg('count').to_frame().reset_index()
df_route['Pct'] = round(df_route['trip_uuid']*100/ df_route['trip_uuid'].sum(),2)
df_route
```

	route_type	trip_uuid	Pct
0	Carting	8908	60.12
1	FTL	5909	39.88

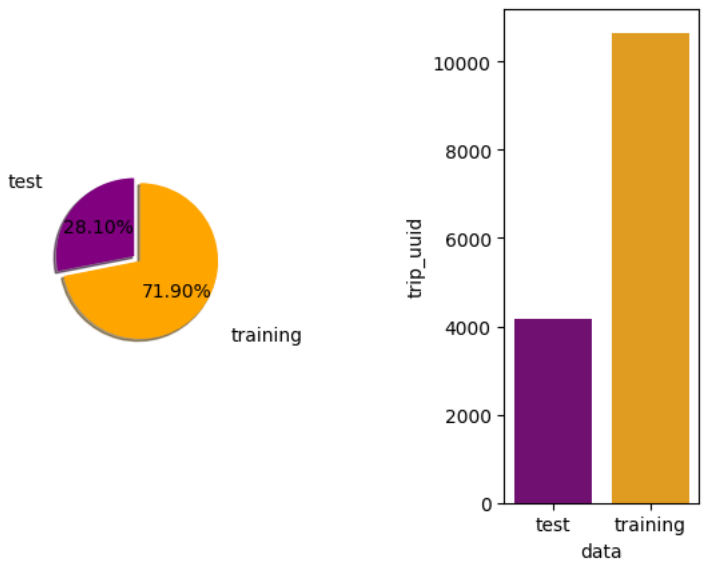
```
# Pie chart
plt.subplot(1, 3, 1)  # Create a subplot for the pie chart
plt.pie(x = df_route['trip_uuid'],colors = ['purple','orange'] ,explode = [0,0.1],autopct = '%.2f%%',labels = ['Carting','FTL'],shadow=True,

# Bar chart
plt.subplot(1, 3, 3)  # Create a subplot for the bar chart
sns.barplot(data = df_route , x= df_route['route_type'], y = df_route['trip_uuid'])

plt.show()
```




```
# Distribution of the data by orders:
df_data = df1.groupby('data')['trip_uuid'].agg('count').to_frame().reset_index()
df_data['pct'] = round((df_data['trip_uuid'] * 100)/df_data['trip_uuid'].sum(),2)
df_data
```

	data	trip_uuid	pct
0	test	4163	28.1
1	training	10654	71.9

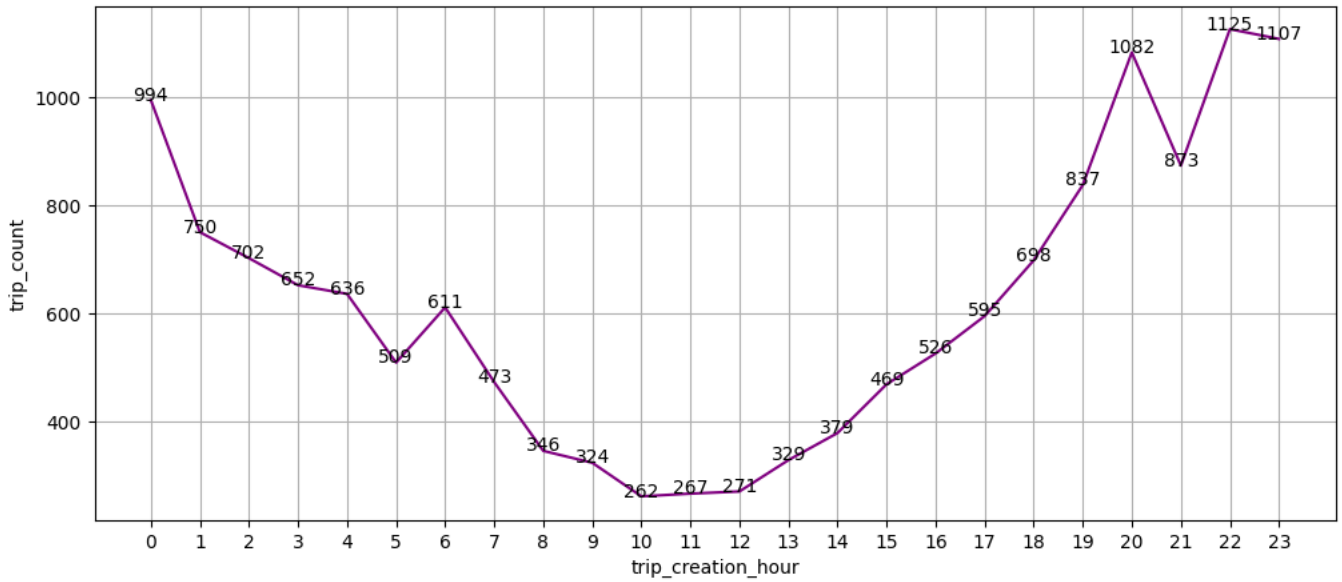
```
# Pie chart
plt.subplot(1, 3, 1)  # Create a subplot for the pie chart
plt.pie(x=df_data['trip_uuid'], colors=['purple', 'orange'], explode=[0, 0.1], autopct='%.2f%%', labels=df_data['data'], shadow=True,labeldi

# Bar chart
plt.subplot(1, 3, 3)  # Create a subplot for the bar chart
sns.barplot(data = df_data, x=df_data['data'],y= df_data['trip_uuid'], palette=['purple', 'orange'])

plt.show()
```




```
# How many trips are created on the hourly basis?
df_hour = df1.groupby('trip_creation_hour')['trip_uuid'].agg('count').to_frame().reset_index()
df_hour.rename(columns = {'trip_uuid':'trip_count'},inplace =True)
df_hour.head()
```

	trip_creation_hour	trip_count
0	0	994
1	1	750
2	2	702
3	3	652
4	4	636

```
plt.figure(figsize = (12,5))
sns.lineplot(data = df_hour,x = df_hour['trip_creation_hour'],y = df_hour['trip_count'],color="purple")
plt.xticks(np.arange(0,24))
addlabels(df_hour['trip_creation_hour'],df_hour['trip_count'])
plt.grid()
plt.show()
```

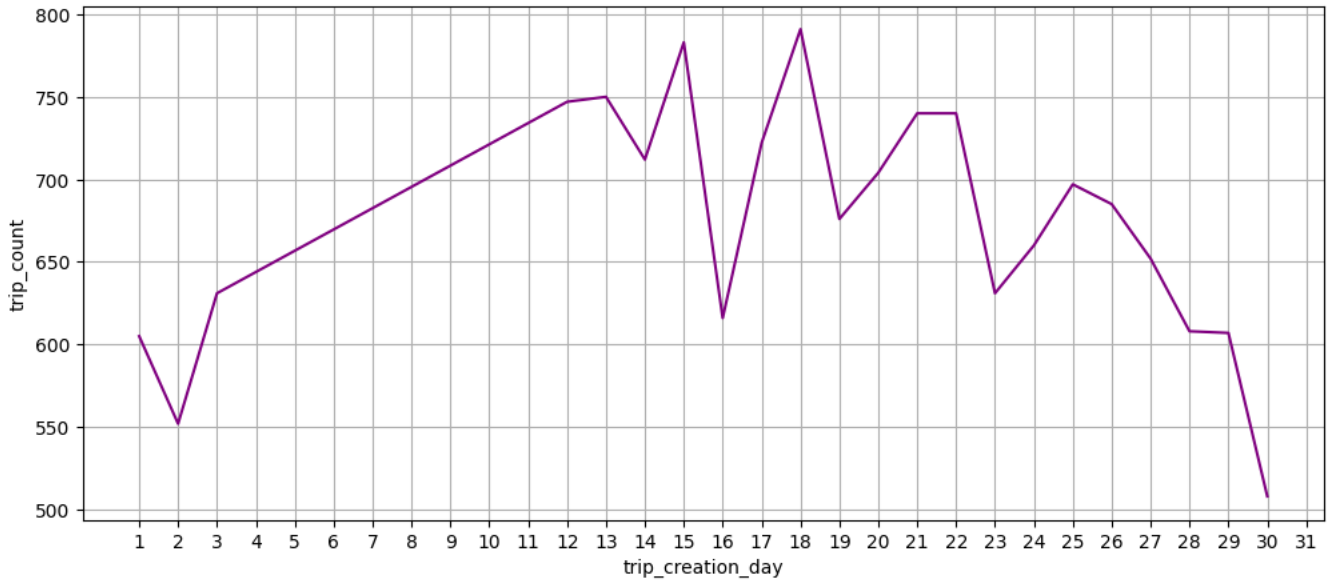


- It can be inferred from the above plot that the number of trips start increasing at afternoon drastically.
- Peak hour is at 10 P.M and then start decreasing.
- And not peek hour is at 5 Am in early morning.

```
# How many trips are created for different days of the month?
df_day = df1.groupby('trip_creation_day')['trip_uuid'].agg('count').to_frame().reset_index()
df_day.rename(columns= {'trip_uuid':'trip_count'},inplace=True)
df_day.head()
```

	trip_creation_day	trip_count
0	1	605
1	2	552
2	3	631
3	12	747
4	13	750

```
plt.figure(figsize = (12,5))
sns.lineplot(data = df_day, x = df_day['trip_creation_day'],y = df_day['trip_count'],color='purple')
plt.xticks(np.arange(1,32))
plt.grid()
plt.show()
```

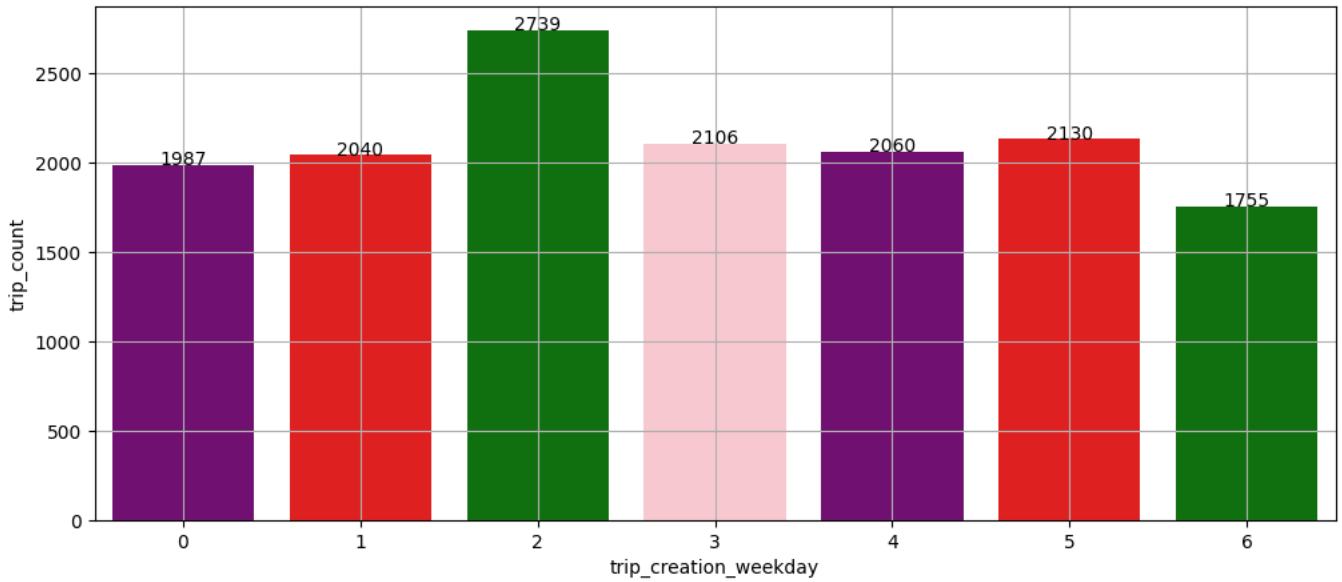


- We can infer that customer make orders usually in the mid of the month most of month.
- And less number of order in the stat and end of the month.

```
# how many trips were created in which day?
df_week_day = df1.groupby('trip_creation_weekday')['trip_uuid'].agg('count').to_frame().reset_index()
df_week_day.rename(columns = {'trip_uuid':'trip_count'},inplace = True)
df_week_day
```

	trip_creation_weekday	trip_count
0	0	1987
1	1	2040
2	2	2739
3	3	2106
4	4	2060
5	5	2130
6	6	1755

```
plt.figure(figsize = (12,5))
sns.barplot(data=df_week_day, x=df_week_day['trip_creation_weekday'], y=df_week_day['trip_count'], palette=['purple', 'red', 'green', 'pink'
addlabels(df_week_day['trip_creation_weekday'], df_week_day['trip_count'])
plt.grid()
plt.show()
```

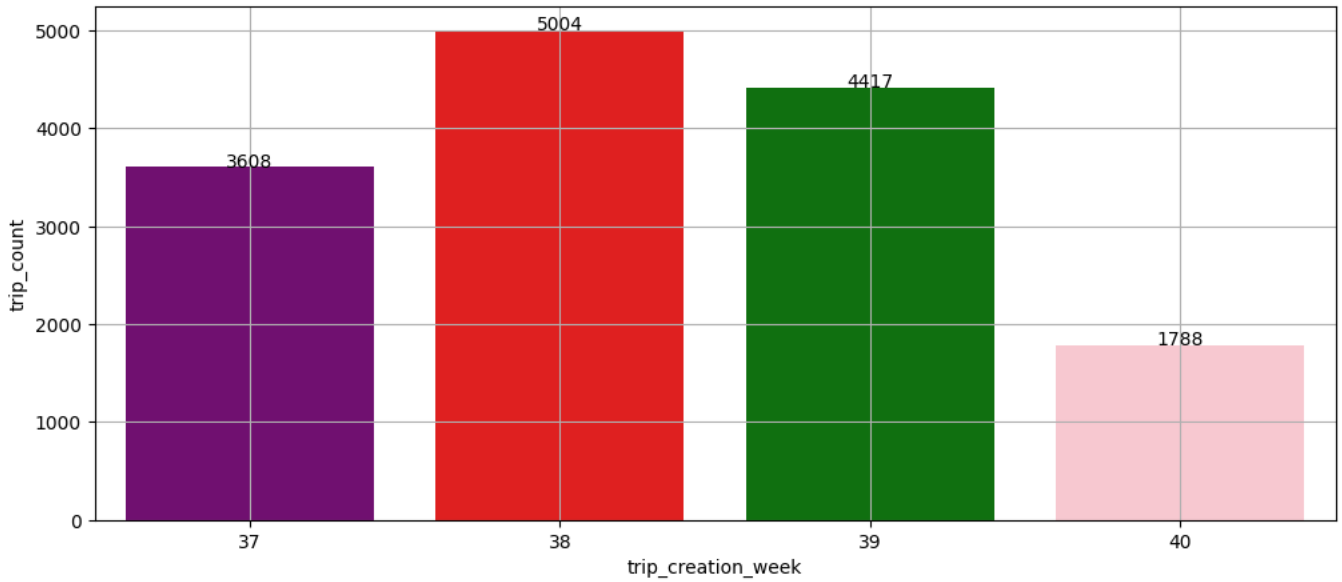


Tuesday is the day with the highest trip creation time. This could be because that there is a day gap given post the weekend when people would make the most purchases so as to consolidate the orders and reduce trips.

```
# How many trips were created in the week?
df_week = df1.groupby('trip_creation_week')['trip_uuid'].agg('count').to_frame().reset_index()
df_week.rename(columns = {'trip_uuid':'trip_count'},inplace=True)
df_week.head()
```

	trip_creation_week	trip_count
0	37	3608
1	38	5004
2	39	4417
3	40	1788

```
plt.figure(figsize = (12,5))
#sns.lineplot(data = df_week, x = df_week['trip_creation_week'],y = df_week['trip_count'],color='purple')
sns.barplot(data=df_week, x=df_week['trip_creation_week'], y=df_week['trip_count'], palette=['purple', 'red', 'green', 'pink'])
addlabels(df_week['trip_creation_week'], df_week['trip_count'])
plt.grid()
plt.show()
```

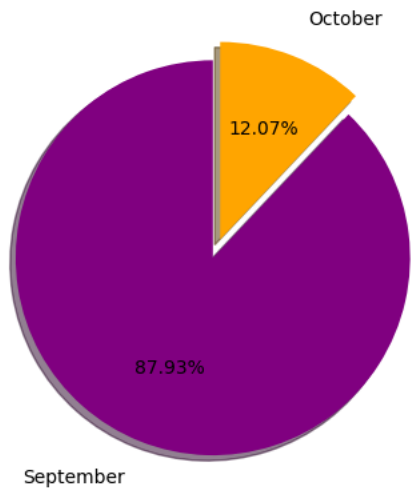


- We can infer from the above plot that most of the trips were made in the 38th week

```
# How many trips are created in the given two months?
df_month = df1.groupby(by = 'trip_creation_month')['trip_uuid'].count().to_frame().reset_index()
df_month['pct'] = np.round(df_month['trip_uuid'] * 100/ df_month['trip_uuid'].sum(), 2)
df_month.head()
```

	trip_creation_month	trip_uuid	pct
0	9	13029	87.93
1	10	1788	12.07

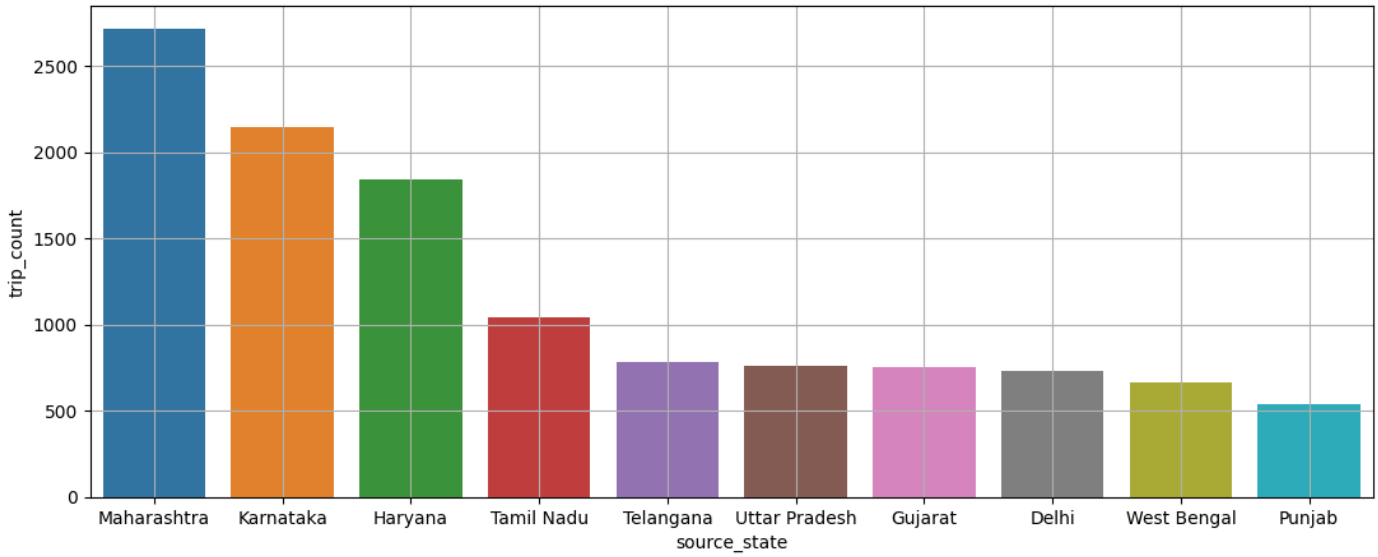
```
# Create a subplot for the pie chart
plt.pie(x=df_month['trip_uuid'], colors=['purple', 'orange'], explode=[0, 0.1], autopct='%.2f%%', labels = ['September','October'], shadow=T
plt.show()
```




```
# How many top 10 number of trips made from different states?
df_state = df1.groupby('source_state')['trip_uuid'].count().to_frame().reset_index()
df_state.rename(columns={'trip_uuid':'trip_count'},inplace = True)
df_state = df_state.sort_values(by='trip_count',ascending = False)[:10]
df_state
```

	source_state	trip_count
17	Maharashtra	2714
14	Karnataka	2143
10	Haryana	1838
24	Tamil Nadu	1039
25	Telangana	781
26	Uttar Pradesh	762
9	Gujarat	750
7	Delhi	728
28	West Bengal	665
22	Punjab	536

```
plt.figure(figsize = (13, 5))
sns.barplot(data = df_state, y = df_state['trip_count'], x = df_state['source_state'])
plt.grid()
plt.show()
```



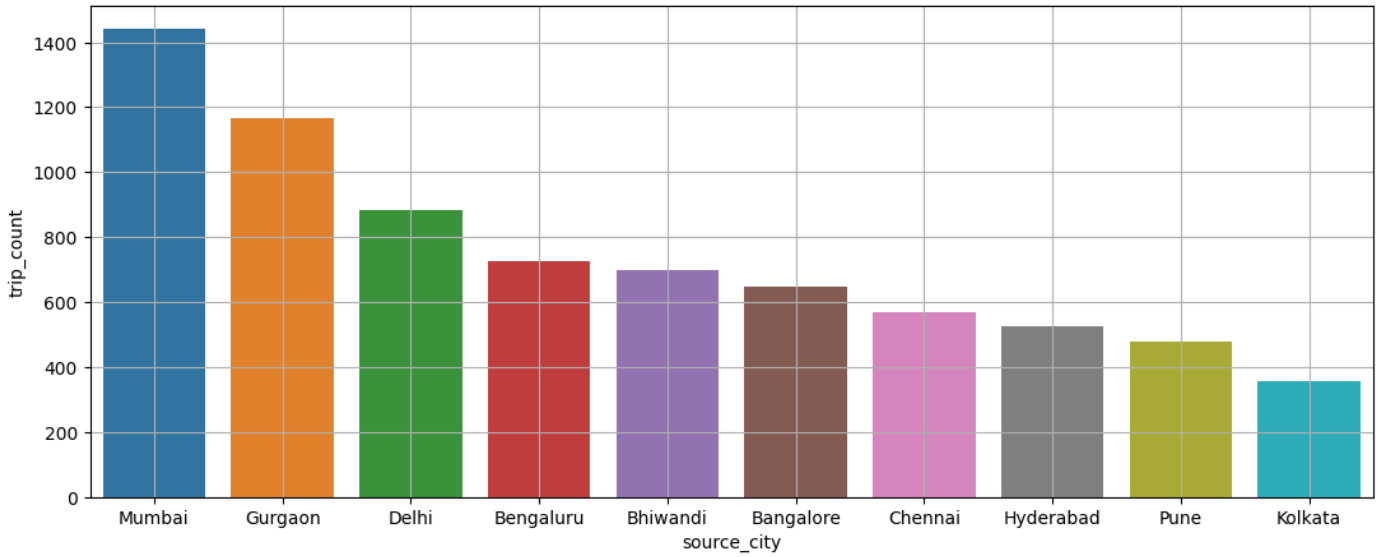
CONCLUSION:

From the above plot we can observe that maximum trips originated from Maharashtra state followed by Karnataka and Haryana. That means that the seller base is strong in these states.

```
# Top 10 number of trips made from different cities?
df_city = df1.groupby('source_city')['trip_uuid'].count().to_frame().reset_index()
df_city.rename(columns={'trip_uuid':'trip_count'},inplace = True)
df_city = df_city.sort_values(by='trip_count',ascending = False)[:10]
df_city
```

	source_city	trip_count
439	Mumbai	1442
237	Gurgaon	1165
169	Delhi	883
79	Bengaluru	726
100	Bhiwandi	697
58	Bangalore	648
136	Chennai	568
264	Hyderabad	524
516	Pune	480
357	Kolkata	356

```
plt.figure(figsize = (13, 5))
sns.barplot(data = df_city, y = df_city['trip_count'], x = df_city['source_city'])
plt.grid()
plt.show()
```



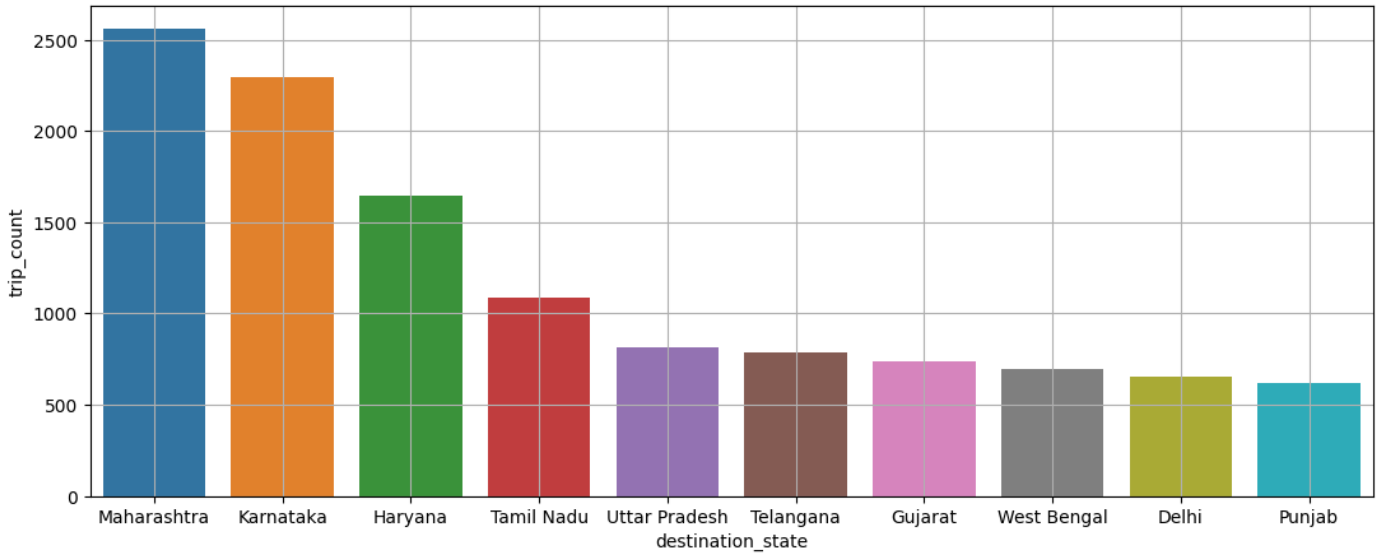
CONCLUSION:

From the above plot we can observe that maximum trips originated from Mumbai state followed by Gurgoan and Delhi. That means that the seller base is strong in these cities.

```
# Top 10 states where number of trips were ended there?
df_des_state = df1.groupby('destination_state')['trip_uuid'].count().to_frame().reset_index()
df_des_state.rename(columns={'trip_uuid':'trip_count'},inplace = True)
df_des_state = df_des_state.sort_values(by='trip_count',ascending = False)[:10]
df_des_state
```

	destination_state	trip_count
18	Maharashtra	2561
15	Karnataka	2294
11	Haryana	1643
25	Tamil Nadu	1084
28	Uttar Pradesh	811
26	Telangana	784
10	Gujarat	734
30	West Bengal	697
8	Delhi	652
23	Punjab	617

```
plt.figure(figsize = (13, 5))
sns.barplot(data = df_des_state, y = df_des_state['trip_count'], x = df_des_state['destination_state'])
plt.grid()
plt.show()
```



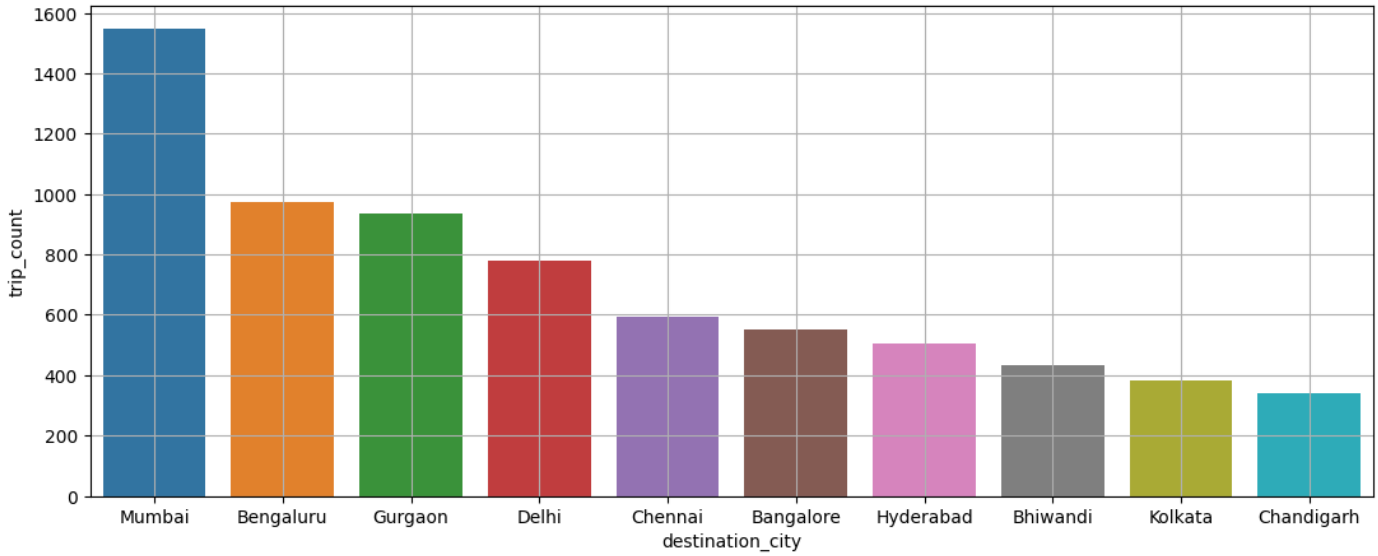
CONCLUSION:

From the above plot we can observe that maximum trips ended at Maharashtra state followed by Karnataka and Haryana. That means that the highest demands are strong in these states.

```
# Top 10 city where number of trips were ended?
df_des_city = df1.groupby('destination_city')['trip_uuid'].count().to_frame().reset_index()
df_des_city.rename(columns={'trip_uuid':'trip_count'},inplace = True)
df_des_city = df_des_city.sort_values(by='trip_count',ascending = False)[:10]
df_des_city
```

	destination_city	trip_count
515	Mumbai	1548
96	Bengaluru	975
282	Gurgaon	936
200	Delhi	778
163	Chennai	595
72	Bangalore	551
308	Hyderabad	503
115	Bhiwandi	434
418	Kolkata	384
158	Chandigarh	339

```
plt.figure(figsize = (13, 5))
sns.barplot(data = df_des_city, y = df_des_city['trip_count'], x = df_des_city['destination_city'])
plt.grid()
plt.show()
```



CONCLUSION:

From the above plot we can observe that maximum trips ended at Mumbai city followed by Bengaluru and Gurgaon. That means that the demands are strong in these cities.

Outlier Detection & Treatment:

a. Find any existing outliers in numerical features.

b. Visualize the outlier values using Boxplot.

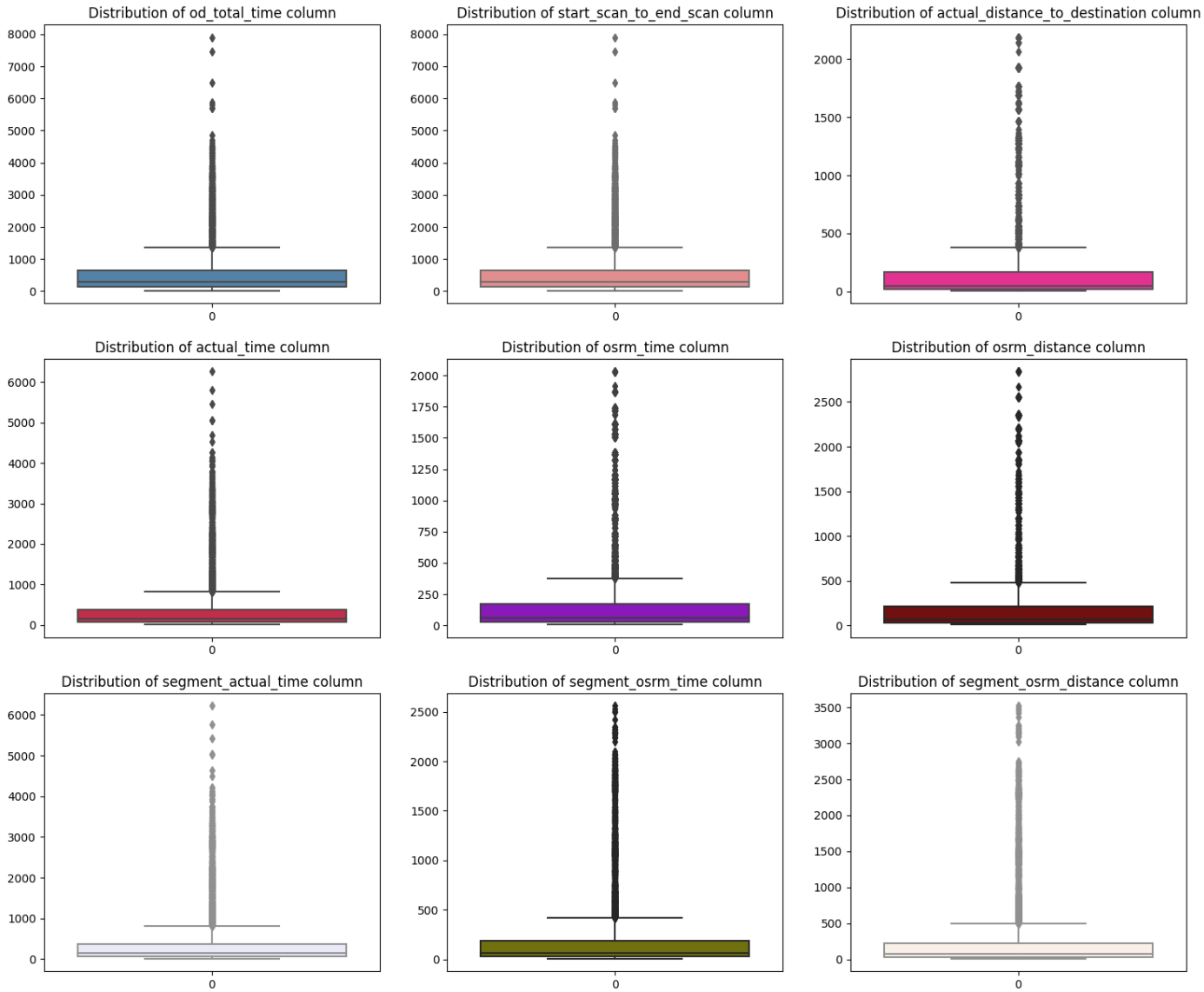
c. Handle the outliers using the IQR method.

```
numerical_columns = ['od_total_time', 'start_scan_to_end_scan', 'actual_distance_to_destination','actual_time', 'osrm_time', 'osrm_distance',
                     'segment_osrm_time', 'segment_osrm_distance']
outliers = df1[numerical_columns].describe().T[['mean','50%']]
outliers['difference'] = outliers['mean'] - outliers['50%']
outliers = outliers[outliers['difference'] > 0]
pd.DataFrame(outliers)
```

	mean	50%	difference
od_total_time	531.795209	280.765626	251.029583
start_scan_to_end_scan	530.810016	280.000000	250.810016
actual_distance_to_destination	164.477829	48.474072	116.003757
actual_time	357.143768	149.000000	208.143768
osrm_time	161.384018	60.000000	101.384018
osrm_distance	204.344711	65.618805	138.725906
segment_actual_time	353.892273	147.000000	206.892273
segment_osrm_time	180.949783	65.000000	115.949783
segment_osrm_distance	223.201157	70.154404	153.046753

There in the above columns we can see that there are potential outliers presernt in the columns

```
# Lets visualize it:
plt.figure(figsize = (18, 15))
for i in range(len(numerical_columns)):
    plt.subplot(3, 3, i + 1)
    clr = np.random.choice(list(mpl.colors.cnames))
    sns.boxplot(df1[numerical_columns[i]], color = clr)
    plt.title(f"Distribution of {numerical_columns[i]} column")
    plt.plot()
```



OUTLIER HANDLING USING IQR

```
# calculate q1(25th percentile) and q3 (75th percentile)
Q1 = df1[numerical_columns].quantile(0.25)
Q3 = df1[numerical_columns].quantile(0.75)
# IQR is our Inter quartile Range
IQR = Q3 - Q1

# Filtering out the outliers by keeping only the values that are within 1.5*IQR of Q1 and Q3
df1_no_outliers = df1[~((df1[numerical_columns] < (Q1 - 1.5 * IQR)) | (df1[numerical_columns] > (Q3 + 1.5 * IQR))).any(axis=1)]

# Comparing the shape of the original and outlier-removed dataframes
original_shape = df1.shape
outlier_removed_shape = df1_no_outliers.shape

original_shape, outlier_removed_shape
```

```
((14817, 28), (12759, 28))
```

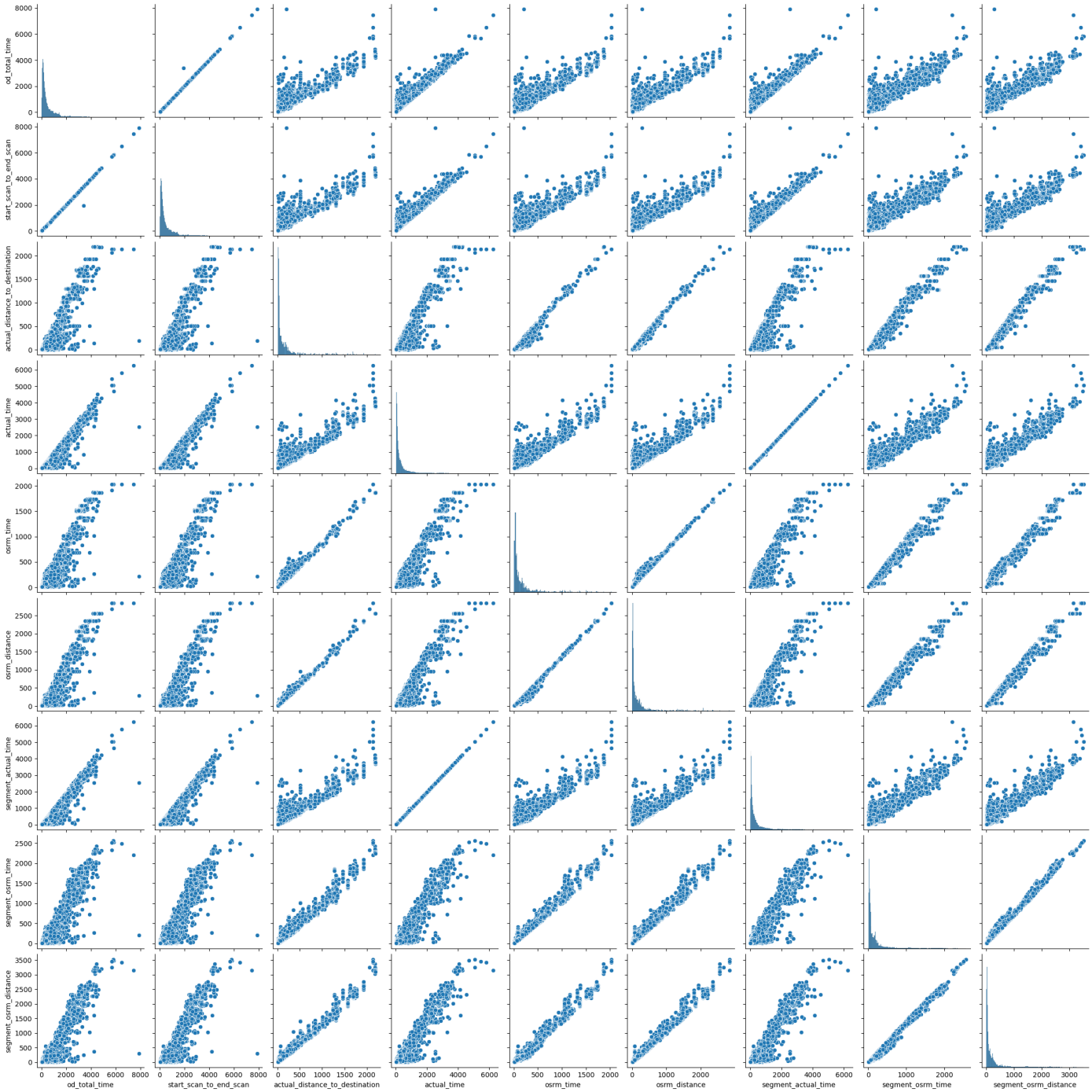
```
# Lets calculate number of outlier present in each features:
for i in numerical_columns:
  q1 = np.quantile(df1[i],0.25)
  q3 = np.quantile(df1[i],0.75)
  IQR = q3 - q1
  lower_bracket = q1 - 1.5 * IQR
  upper_bracket = q3 -  1.5 * IQR
  cal_outlier = df1.loc[(df1[i]<lower_bracket) | df1[i]>upper_bracket]
  print('COLUMN:', i)
  print(f'Q1 : {q1}')
  print(f'Q3 : {q3}')
  print(f'IQR : {IQR}')
  print(f'LOWER BRACKET : {lower_bracket}')
  print(f'UPPER BRACKET : {upper_bracket}')
  print(f'NUMBER OF OUTLIERS : {cal_outlier.shape[0]}')
  print('--'*20)
```

```
NUMBER OF OUTLIERS : 14817
----------------------------------------
COLUMN: actual_distance_to_destination
Q1 : 22.837238311767578
Q3 : 164.5832061767578
IQR : 141.74596786499023
LOWER BRACKET : -189.78171348571777
UPPER BRACKET : -48.03574562072754
NUMBER OF OUTLIERS : 14817
----------------------------------------
COLUMN: actual_time
Q1 : 67.0
Q3 : 370.0
IQR : 303.0
LOWER BRACKET : -387.5
UPPER BRACKET : -84.5
NUMBER OF OUTLIERS : 14817
----------------------------------------
COLUMN: osrm_time
Q1 : 29.0
Q3 : 168.0
IQR : 139.0
LOWER BRACKET : -179.5
UPPER BRACKET : -40.5
NUMBER OF OUTLIERS : 14817
----------------------------------------
COLUMN: osrm_distance
Q1 : 30.81920051574707
Q3 : 208.47500610351562
IQR : 177.65580558776855
LOWER BRACKET : -235.66450786590576
UPPER BRACKET : -58.00870227813721
```

```
LOWER BRACKET : -200.0
UPPER BRACKET : -46.0
NUMBER OF OUTLIERS : 14817
----------------------------------------
COLUMN: segment_osrm_distance
Q1 : 32.65449905395508
Q3 : 218.80239868164062
IQR : 186.14789962768555
LOWER BRACKET : -246.56735038757324
UPPER BRACKET : -60.419450759887695
NUMBER OF OUTLIERS : 14817
----------------------------------------
```

PAIR PLOT:

```
sns.pairplot(df1[numerical_columns])
plt.show()
```

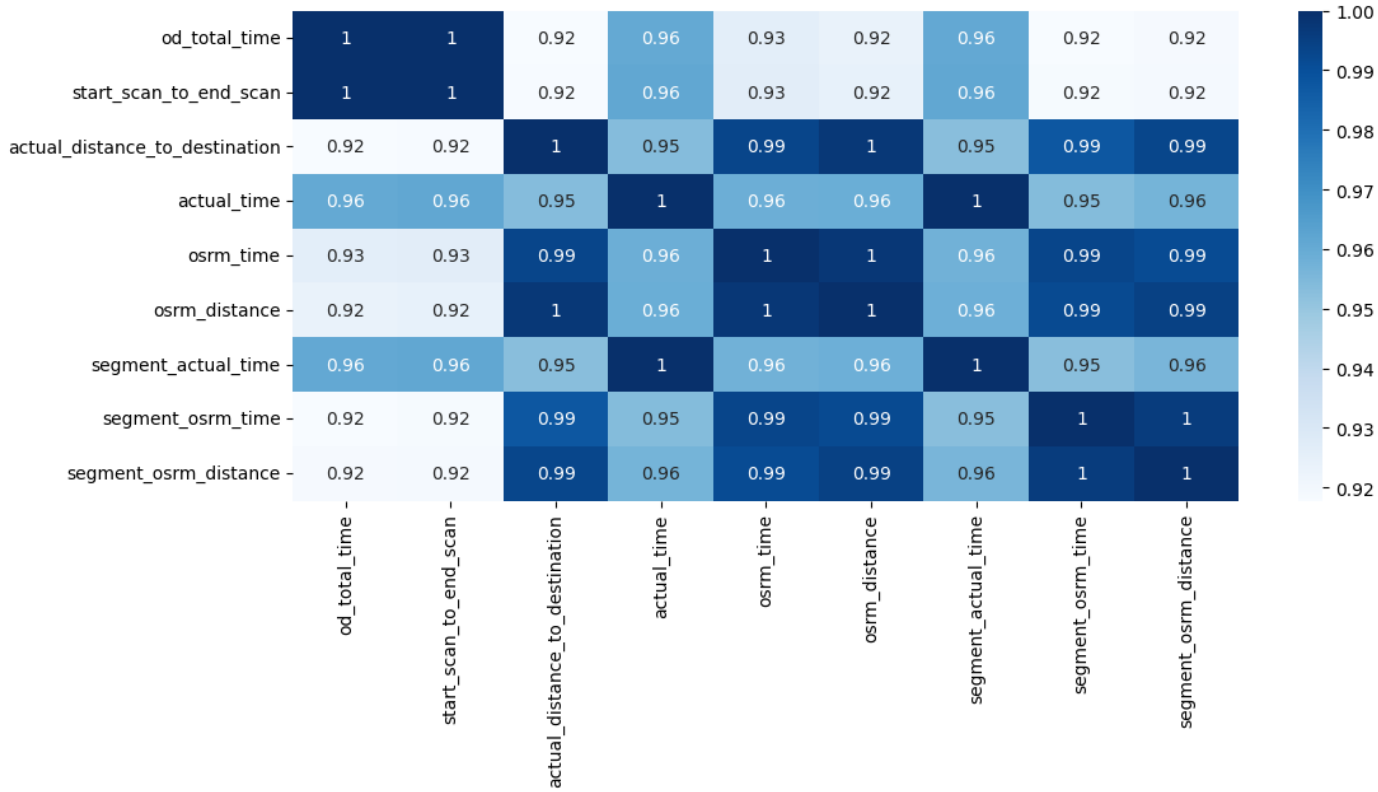


HEAT MAP:

```
df1_corr = df1[numerical_columns].corr()
```

```
plt.figure(figsize = (12, 5))
sns.heatmap(data = df1_corr, annot = True,cmap ='Blues')
plt.plot()
```

```
[]
```



Data are very highly correlated i.e. > 0.99 between columns actual_distance_to_destination and orsm_time, segment_orsm_time and segment_orsm_distance.

Perform one-hot encoding on categorical features

one-hot encodiing for data:

```
# import library:
from sklearn.preprocessing import LabelEncoder
```

```
# Original column:
df1['data'].value_counts()
```

```
training    10654
test         4163
Name: data, dtype: int64
```

```
laber_encoder = LabelEncoder()
df1['data'] = label_encoder.fit_transform(df1['data'])
```

```
df1['data'].value_counts()
```

```
1    10654
0     4163
Name: data, dtype: int64
```

one-hot encodiing for route type:

```
# Original column:
df1['route_type'].value_counts()
```

```
Carting    8908
FTL        5909
Name: route_type, dtype: int64
```

```
# Perform one-hot encoding on categorical column route type:
df1['route_type'] = label_encoder.fit_transform(df1['route_type'])
```

```
df1['route_type'].value_counts()
```

```
0    8908
1    5909
Name: route_type, dtype: int64
```

Normalize/ Standardize the numerical features using MinMaxScaler or StandardScaler

```
# Import library for normalization and standardization:(minmaxscaler and standardscaler)
from sklearn.preprocessing import StandardScaler, MinMaxScaler
```

Standardization:

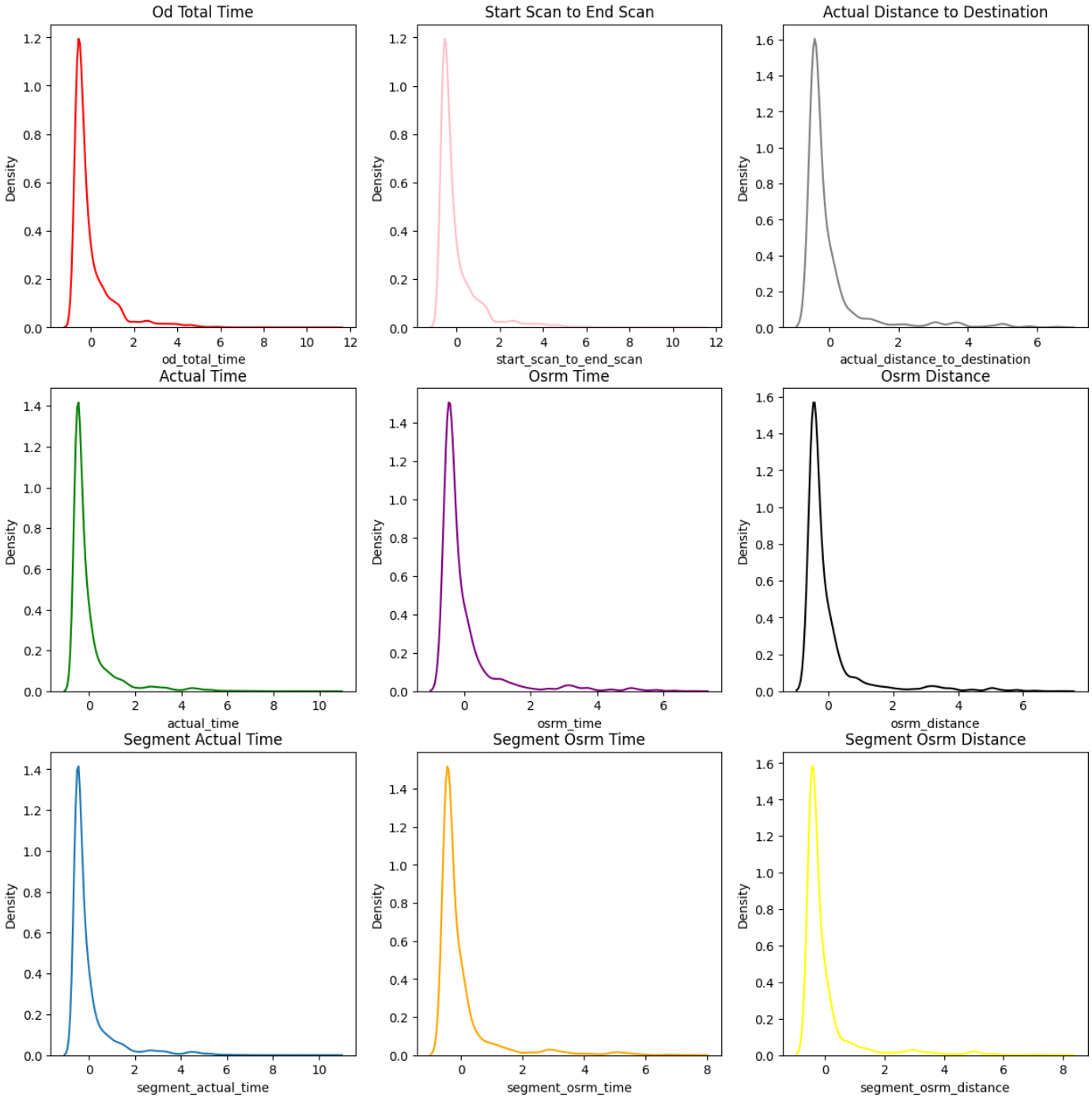
```
numerical_columns
```

```
['od_total_time',
 'start_scan_to_end_scan',
 'actual_distance_to_destination',
 'actual_time',
 'osrm_time',
 'osrm_distance',
 'segment_actual_time',
 'segment_osrm_time',
 'segment_osrm_distance']
```

```
standard_data = StandardScaler().fit_transform(df1[numerical_columns])
df_standard_data =pd.DataFrame(standard_data,columns = numerical_columns)
df_standard_data.head()
```

	od_total_time	start_scan_to_end_scan	actual_distance_to_destination	actual_time	osrm_time	osrm_distance	segment_actual_time	se
0	2.621986	2.623702	2.162092	2.146251	2.047585	2.124848	2.146791	
1	-0.531255	-0.532593	-0.298944	-0.381461	-0.344144	-0.321920	-0.382742	
2	5.161957	5.165134	5.772935	5.325931	5.817598	5.804050	5.310954	
3	-0.654316	-0.654047	-0.482362	-0.531093	-0.539462	-0.498578	-0.530163	
4	0.283017	0.282670	-0.121257	-0.028757	-0.163566	-0.155387	-0.024976	

```
plt.figure(figsize = (15, 15))
plt.subplot(3,3,1)
sns.kdeplot(df_standard_data['od_total_time'],color = 'red')
plt.title('Od Total Time')

plt.subplot(3,3,2)
sns.kdeplot(df_standard_data['start_scan_to_end_scan'],color = 'pink')
plt.title('Start Scan to End Scan')

plt.subplot(3,3,3)
sns.kdeplot(df_standard_data['actual_distance_to_destination'],color = 'grey')
plt.title('Actual Distance to Destination')

plt.subplot(3,3,4)
sns.kdeplot(df_standard_data['actual_time'],color = 'green')
plt.title('Actual Time')

plt.subplot(3,3,5)
sns.kdeplot(df_standard_data['osrm_time'],color = 'purple')
plt.title('Osrm Time')

plt.subplot(3,3,6)
sns.kdeplot(df_standard_data['osrm_distance'],color = 'black')
plt.title('Osrm Distance')

plt.subplot(3,3,7)
sns.kdeplot(df_standard_data['segment_actual_time'])
plt.title('Segment Actual Time')

plt.subplot(3,3,8)
sns.kdeplot(df_standard_data['segment_osrm_time'],color = 'orange')
plt.title('Segment Osrm Time')

plt.subplot(3,3,9)
sns.kdeplot(df_standard_data['segment_osrm_distance'],color = 'yellow')
plt.title('Segment Osrm Distance')

plt.show()
```

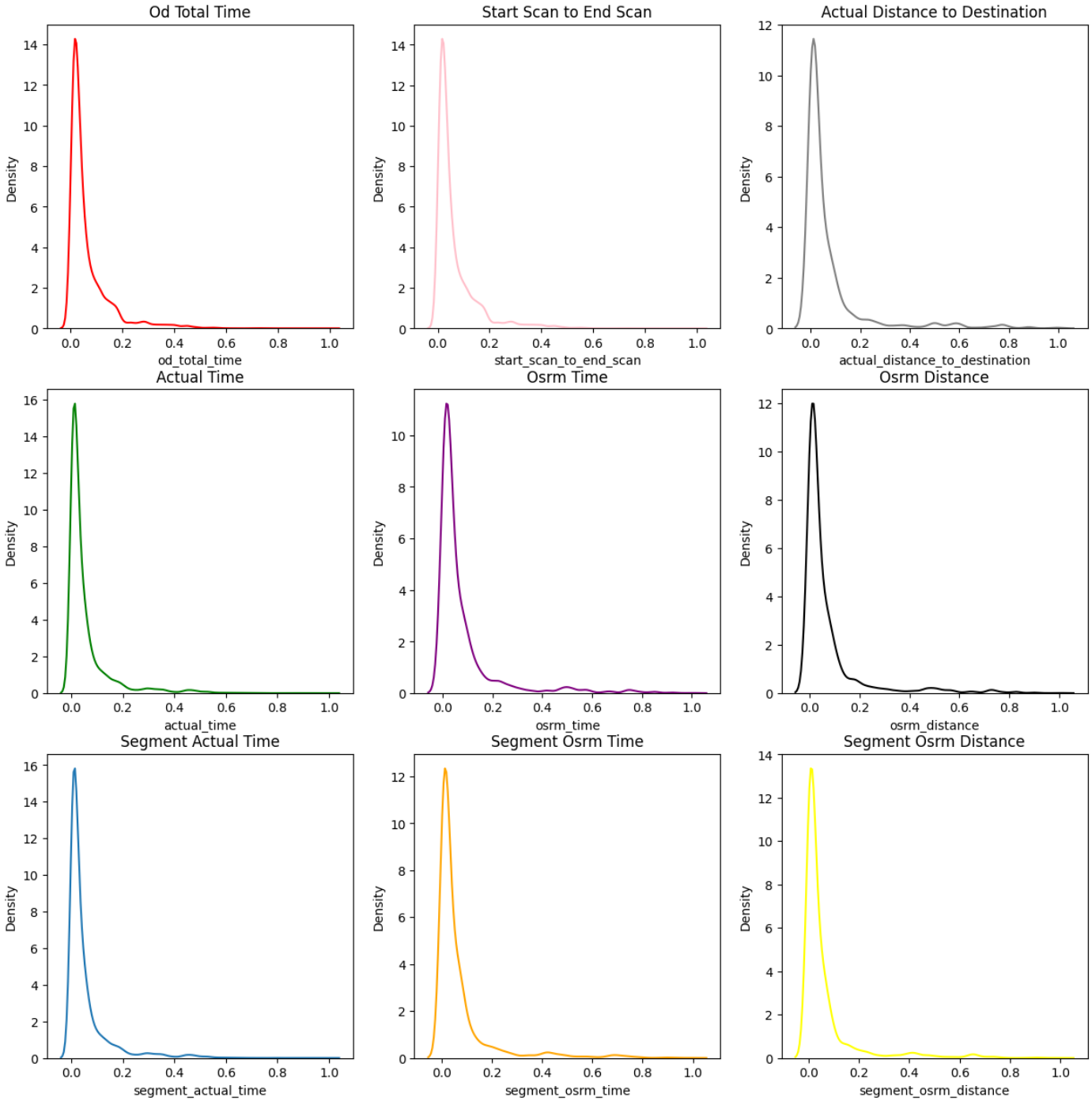


Normalizatioin:

```
normalize_data = MinMaxScaler().fit_transform(df1[numerical_columns])
df_normalized = pd.DataFrame(normalize_data,columns = numerical_columns)
df_normalized.head()
```

	od_total_time	start_scan_to_end_scan	actual_distance_to_destination	actual_time	osrm_time	osrm_distance	segment_actual_time	se
0	0.284016	0.283937	0.374613	0.248242	0.350938	0.346972	0.247388	
1	0.020082	0.019937	0.029476	0.021419	0.030602	0.026859	0.021218	
2	0.496617	0.496508	0.880999	0.533568	0.855874	0.828325	0.530301	
3	0.009782	0.009778	0.003753	0.007992	0.004442	0.003747	0.008037	
4	0.088239	0.088127	0.054395	0.053069	0.054788	0.048647	0.053207	

```
plt.figure(figsize = (15, 15))
plt.subplot(3,3,1)
sns.kdeplot(df_normalized['od_total_time'],color = 'red')
plt.title('Od Total Time')

plt.subplot(3,3,2)
sns.kdeplot(df_normalized['start_scan_to_end_scan'],color = 'pink')
plt.title('Start Scan to End Scan')

plt.subplot(3,3,3)
sns.kdeplot(df_normalized['actual_distance_to_destination'],color = 'grey')
plt.title('Actual Distance to Destination')

plt.subplot(3,3,4)
sns.kdeplot(df_normalized['actual_time'],color = 'green')
plt.title('Actual Time')

plt.subplot(3,3,5)
sns.kdeplot(df_normalized['osrm_time'],color = 'purple')
plt.title('Osrm Time')

plt.subplot(3,3,6)
sns.kdeplot(df_normalized['osrm_distance'],color = 'black')
plt.title('Osrm Distance')

plt.subplot(3,3,7)
sns.kdeplot(df_normalized['segment_actual_time'])
plt.title('Segment Actual Time')

plt.subplot(3,3,8)
sns.kdeplot(df_normalized['segment_osrm_time'],color = 'orange')
plt.title('Segment Osrm Time')

plt.subplot(3,3,9)
sns.kdeplot(df_normalized['segment_osrm_distance'],color = 'yellow')
plt.title('Segment Osrm Distance')

plt.show()
```



By Normalizing out features we can observe that all our data is between the range of 0 to 1.

Hypothesis Testing:

Perform hypothesis testing / visual analysis between : Lets formulate everything:

STEP-1 : Set up Null Hypothesis

Null Hypothesis (H0) - od_total_time (Total Trip Time) and start_scan_to_end_scan (Expected total trip time) are same.

Alternate Hypothesis (HA) - od_total_time (Total Trip Time) and start_scan_to_end_scan (Expected total trip time) are different.

STEP-2 : Checking for basic assumpitons for the hypothesis

Distribution check using QQ Plot Homogeneity of Variances using Lavene's test

STEP-3: Define Test statistics; Distribution of T under H0.

If the assumptions of T Test are met then we can proceed performing T Test for independent samples else we will perform the non parametric test equivalent to T Test for independent sample i.e., Mann-Whitney U rank test for two independent samples.

STEP-4: Compute the p-value and fix value of alpha.

(We set our *alpha to be 0.05*)

STEP-5: Compare p-value and alpha.

Based on p-value, we will accept or reject H0.

p-val > alpha : Accept H0 p-val < alpha : Reject H0

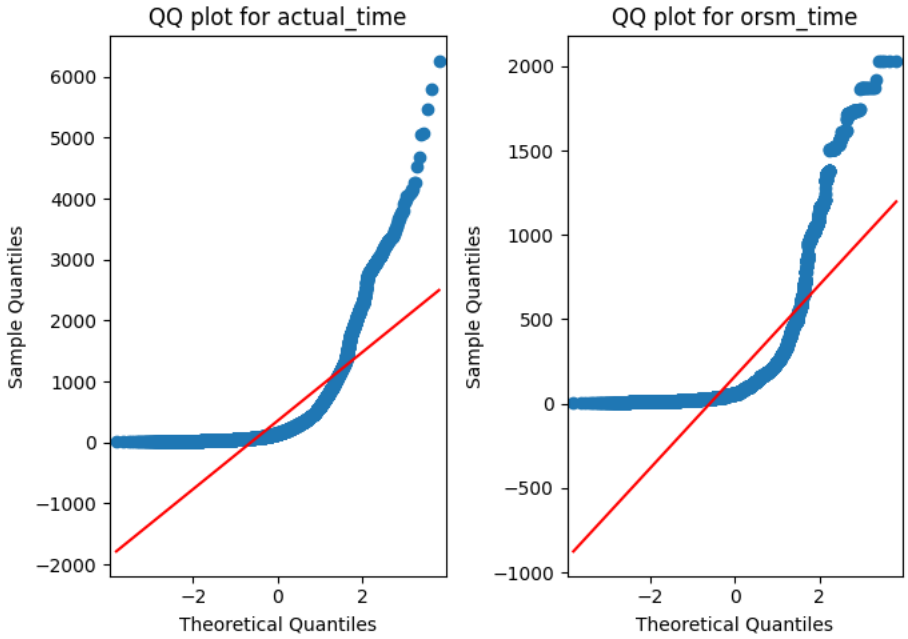
```
# Import all the necessary library:
from statsmodels.graphics.gofplots import qqplot
from scipy.stats import kruskal, shapiro, levene, boxcox,kruskal
```

a. Actual_time aggregated value and OSRM time aggregated value.

```
df1[['actual_time','osrm_time']].describe()
```

	actual_time	osrm_time
count	14817.000000	14817.000000
mean	357.143768	161.384018
std	561.396118	271.360992
min	9.000000	6.000000
25%	67.000000	29.000000
50%	149.000000	60.000000
75%	370.000000	168.000000
max	6265.000000	2032.000000

```
# Visual test do check for normality of the data:
fig, axes = plt.subplots(1, 2, figsize=(7, 5))

# Plot for actual_time
axes[0].set_title('QQ plot for actual_time')
qqplot(df1['actual_time'], line='s', ax=axes[0])

# Plot for segment_actual_time
axes[1].set_title('QQ plot for orsm_time')
qqplot(df1['osrm_time'], line='s', ax=axes[1])

plt.tight_layout()
plt.show()
```



Conclusion: It can be seen from the above that the sample do not follow noraml distribution

Shapiro test:

```
#H0: The sample data follows normal distribution
#H0: The sample data does not follow normal distribution
test_stat, p_value = shapiro(df1['actual_time'].sample(5000))
print('p_value',p_value)
if p_value < 0.5:
  print('The data does not follow normal distribution')
else:
  print('The data does follows the Normal distribution')
```

```
p_value 0.0
The data does not follow normal distribution
```

```
test_stat, p_value = shapiro(df1['osrm_time'].sample(5000))
print('p-value', p_value)
if p_value < 0.05:
    print('The sample does not follow normal distribution')
else:
    print('The sample follows normal distribution')
```

```
p-value 0.0
The sample does not follow normal distribution
```

Therefore data does not follows the Normal Distribution , so we perform boccox transformation.

```
transformed_actual_time = boxcox(df1['actual_time'])[0]
test_stat, p_value = spy.shapiro(transformed_actual_time)
print('p-value', p_value)
if p_value < 0.05:
    print('The sample does not follow normal distribution')
else:
    print('The sample follows normal distribution')
```

```
p-value 1.020620453603145e-28
The sample does not follow normal distribution
/usr/local/lib/python3.10/dist-packages/scipy/stats/_morestats.py:1882: UserWarning: p-value may not be accurate for N > 5000.
  warnings.warn("p-value may not be accurate for N > 5000.")
```

```
transformed_actual_time = boxcox(df1['osrm_time'])[0]
test_stat, p_value = shapiro(transformed_actual_time)
print('p-value', p_value)
if p_value < 0.05:
    print('The sample does not follow normal distribution')
else:
    print('The sample follows normal distribution')
```

```
p-value 3.5882550510138333e-35
The sample does not follow normal distribution
```

- Even after applying the boxcox transformation on each of the "actual_time" and "osrm_time" columns, the distributions do not follow normal distribution.

Levene test for checking the variances between the 2 groups

```
# H0 ; Data have equal Variance
# HA : Data do not have equal Variance

test_stat, p_value = levene(df1['actual_time'], df1['osrm_time'])
print('p-value', p_value)
if p_value < 0.05:
    print('The samples do not have  Homogenous Variance')
else:
    print('The samples have Homogenous Variance ')
```

```
p-value 1.871098057987424e-220
The samples do not have  Homogenous Variance
```

- Since the samples do not follow any of the assumptions T-Test cannot be applied here, we can perform its non parametric equivalent test i.e., Kruskal test for two independent samples.

```
#Null Hypothesis (H0): There is no significant difference among the medians of the groups.
# Alternative Hypothesis (Ha): At least one group has a different median than the others.
test_stat, p_value = kruskal(df1['actual_time'], df1['osrm_time'])
print('p-value', p_value)
if p_value < 0.05:
    print('The samples are not similar')
else:
    print('The samples are similar ')
```

```
p-value 0.0
The samples are not similar
```

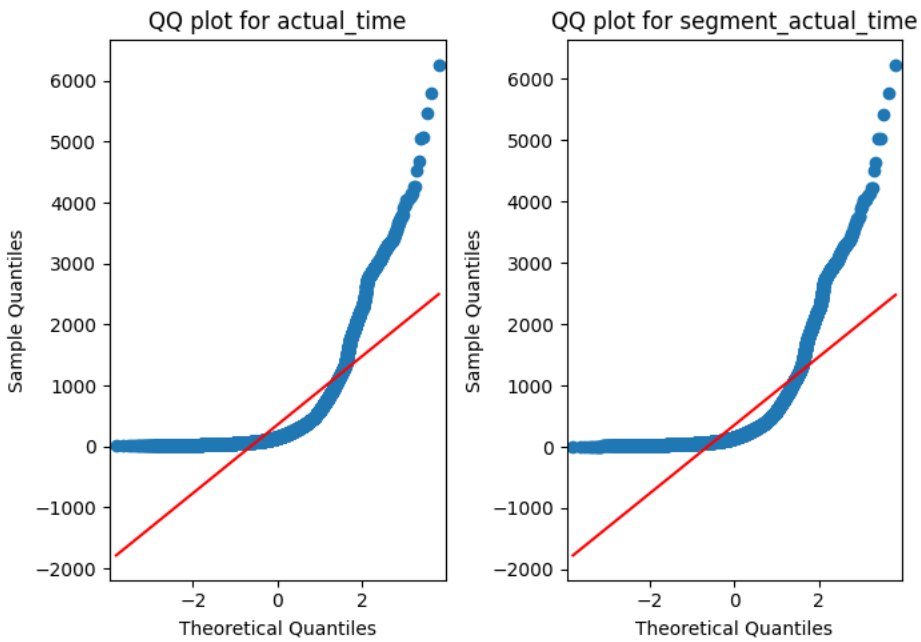
- Therefore final conclusion is that the actual time and osrm time are not similar.

b.Actual_time aggregated value and segment actual time aggregated value.

```
df1[['actual_time', 'segment_actual_time']].describe()
```

	actual_time	segment_actual_time
count	14817.000000	14817.000000
mean	357.143768	353.892273
std	561.396118	556.247925
min	9.000000	9.000000
25%	67.000000	66.000000
50%	149.000000	147.000000
75%	370.000000	367.000000
max	6265.000000	6230.000000

```
# check for the nomality visualize:
fig, axes = plt.subplots(1, 2, figsize=(7, 5))

# Plot for actual_time
axes[0].set_title('QQ plot for actual_time')
qqplot(df1['actual_time'], line='s', ax=axes[0])

# Plot for segment_actual_time
axes[1].set_title('QQ plot for segment_actual_time')
qqplot(df1['segment_actual_time'], line='s', ax=axes[1])

plt.tight_layout()
plt.show()
```



It can be seen from the above plots that the samples do not come from normal distribution.

```
# Check wiith shapiro test:
alpha = 0.05
# H0 : The sample follows normal distribution or gaussian
# Ha : The sample does not follow normal distribution
test_stat, p_value = shapiro(df1['actual_time'].sample(5000))
print('p-value', p_value)
if p_value < 0.05:
    print('The sample does not follow normal distribution')
else:
    print('The sample follows normal distribution')
```

```
p-value 0.0
The sample does not follow normal distribution
```

```
test_stat, p_value = shapiro(df1['segment_actual_time'].sample(5000))
print('p-value', p_value)
if p_value < 0.05:
    print('The sample does not follow normal distribution')
else:
    print('The sample follows normal distribution')
```

```
p-value 0.0
The sample does not follow normal distribution
```

```
# Transform data using boxcox transformation to check if transform data follows normal distribution or not.
transformed_actual_time = boxcox(df1['actual_time'])[0]
test_stat,p_value = shapiro(transformed_actual_time)
print('p-value:', p_value)
if p_value< 0.05:
  print('Sample does not follow normal distribution')
else:
  print('Sample follows normal distribution')
```

```
p-value: 1.020620453603145e-28
Sample does not follow normal distribution
/usr/local/lib/python3.10/dist-packages/scipy/stats/_morestats.py:1882: UserWarning: p-value may not be accurate for N > 5000.
  warnings.warn("p-value may not be accurate for N > 5000.")
```

```
transformed_seg_actual_time = boxcox(df1['segment_actual_time'])[0]
test_stat,p_value = shapiro(transformed_seg_actual_time)
print('p-value:', p_value)
if p_value< 0.05:
  print('Sample does not follow normal distribution')
else:
  print('Sample follows normal distribution')
```

```
p-value: 5.700074948787037e-29
Sample does not follow normal distribution
```

CONCLUSION: Even after applying the boxcox transformation on each of the features the disritbution does follows normal distribution.

```
# Lavene's test checks the Homogeneity of Variances;
# H0: Varaincs is equal between the groups
# Ha: Varaincs is not equal between the groups

test_stat, p_value = levene(df1['actual_time'], df1['segment_actual_time'])
print('p-value', p_value)
if p_value <0.05:
  print('The samples does not have homogenous variances')
else:
  print('The samples have Homogenous Variance')
```

```
p-value 0.695502241317651
The samples have Homogenous Variance
```

Since the samples are not from normal distribution T-Test cannot be applied here, we can perform its non parametric equivalent test i.e., kruskal test for two independent samples.

Kruska test(as all our assumption gets rejected)

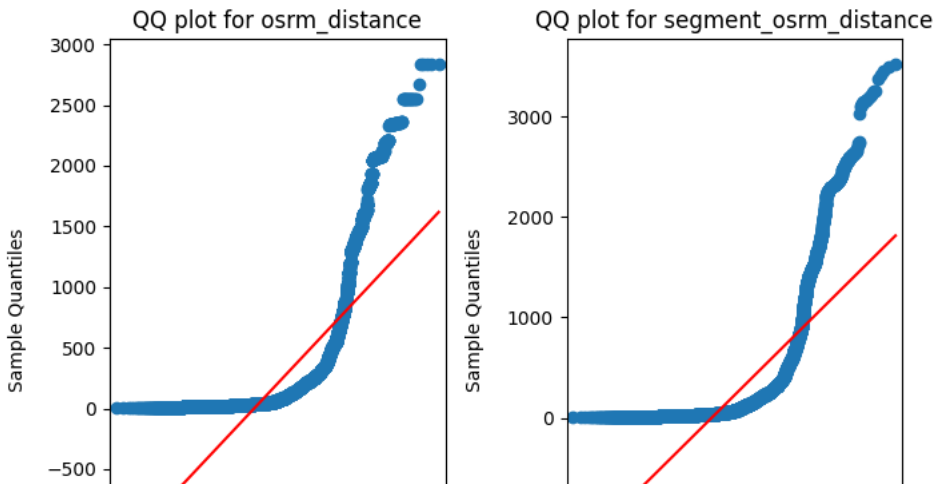
```
# Null hypothesis (H0): The medians of all groups are equal.
# Alternative hypothesis (H1): At least one group's median is different from the others.
test_stat, p_value = kruskal(df1['actual_time'], df1['segment_actual_time'])
print('p-value', p_value)
if p_value < 0.05:
    print('The samples are not similar')
else:
    print('The samples are similar ')
```

```
p-value 0.4164231265256002
The samples are similar
```

Therefore from the KRUSKAL test we can conclude that samples are similar

c. OSRM distance aggregated value and segment OSRM distance aggregated value.

```
df1[['osrm_distance', 'segment_osrm_distance']]
```

	osrm_distance	segment_osrm_distance
0	991.352295	1320.473267
1	85.111000	84.189400
2	2354.066650	2545.267822
3	19.680000	19.876600
4	146.791794	146.791901
...	...	...
14812	73.462997	64.855103
14813	16.088200	16.088299

```
# Visual lets check if data follows normal distribution or not
# check for the nomality visualize:
fig, axes = plt.subplots(1, 2, figsize=(7, 5))

# Plot for actual_time
axes[0].set_title('QQ plot for osrm_distance')
qqplot(df1['osrm_distance'], line='s', ax=axes[0])

# Plot for segment_actual_time
axes[1].set_title('QQ plot for segment_osrm_distance')
qqplot(df1['segment_osrm_distance'], line='s', ax=axes[1])

plt.tight_layout()
plt.show()
```